

PSP Database Optimization

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 - DailyBatchJobsProcessor - fx6bn8avj7cp2
 - More Queries to check
 - BatchJob (Module: BaseATFExtractFileProcess)
 - BatchJob (Module: BaseATFExtractFileProcess)
 - BatchJob (Module: BaseATFExtractFileProcess)
- FYI, below queries got changed and no more opportunity for optimization.

Scope

The focus of this document is on identifying opportunities to improve database performance and take actions accordingly by troubleshooting root cause.

Introduction

There are multiple multi-year business goals like Grow Workforce and Equifax, which have quarterly milestones to achieve. This will bring additional data as well as workload to PSP monolith database. To support these business goals, we need to ensure the PSP monolith database is performing well and have enough headroom of resources to accommodate future increases in workload.

Problem statement

There is high usage in ReadIOPS and ReadThroughput observed in monolith database. GREEN

Changes

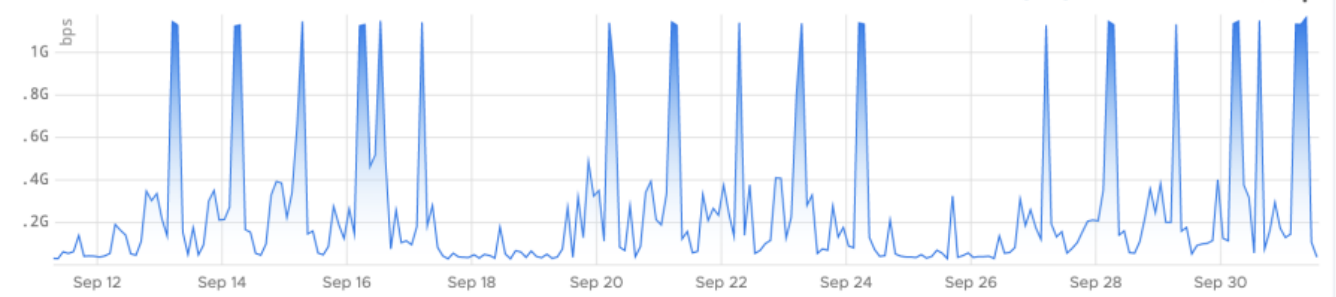
- Queries for sql_ids (3gsxm5sqh6672, 3gv8qjry4fr4z) have been changed.

Opportunity #1

Old state
ReadThroughput (August 2021) => ~790 MB



Current state
ReadThroughput (September 2021) => less than 400 MB



Action Taken
Identified root cause and fixed.

There are still few spikes in ReadThroughput observed.

Current state
ReadThroughput (September 2021) => spikes of 1.1G as shown below

SQL ordered by Reads (24-Sep-21 19:00 - 24-Sep-21 19:45). **Discontinued after 17-Dec-2021**

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	%IO	SQL Id	SQL Module	SQL Text
31,561,112	1	31,561,112.00	83.58	1,147.38	67.17	33.93	6qk524uazdf0c	GemsAccountsReceivableProcesso	UPDATE PSP_FINANCIAL_TRANS_STA...

Opportunity #2

GemsAccountsReceivableProcessor - 6qk524uazdf0c

```

UPDATE PSP_FINANCIAL_TRANS_STATE FTS
SET FTS.GEMS_UPLOAD_BATCH_FK = :B5,
    VERSION                    = VERSION + 1,
    MODIFIER_ID                = :B4,
    MODIFIED_DATE              = :B3
WHERE FTS.GEMS_UPLOAD_BATCH_FK IS NULL
AND TRUNC(FTS.TRANSACTION_STATE_EFF_DATE) >= TRUNC(:B2) - :B1
AND FTS.TRANSACTION_STATE_EFF_DATE >= TRUNC(:B2) - :B1
AND FTS.TRANSACTION_STATE_FK IN ('Executed', 'Voided', 'Completed')
AND EXISTS(SELECT 'T'
            FROM PSP_FINANCIAL_TRANSACTION FT,
                 PSP_POSTING_RULE PR,
                 PSP_LEDGER_ACCOUNT LA
            WHERE FT.FINANCIAL_TRANSACTION_SEQ = FTS.FINANCIAL_TRANSACTION_FK
                  AND PR.TRANSACTION_STATE_FK = FTS.TRANSACTION_STATE_FK
                  AND PR.LEDGER_ACCOUNT_FK = LA.LEDGER_ACCOUNT_CD
                  AND PR.TRANSACTION_TYPE_FK = FT.TRANSACTION_TYPE_FK

```

AND LA.REPORTING_FREQUENCY = 'Daily'
AND LA.LEDGER_ACCOUNT_TYPE IN ('Income', 'SUTax')
AND FT.SKU IS NOT NULL)

Plan hash value: 1845674939

Id	Operation	Name	Rows	Bytes	TempSpc
0	UPDATE STATEMENT				
1	UPDATE	PSP_FINANCIAL_TRANS_STATE			
2	MERGE JOIN		21M	15G	
3	SORT JOIN		128M	76G	
4	VIEW	VW_SQ_1	128M	76G	
5	SORT UNIQUE		128M	15G	16G
6	HASH JOIN		128M	15G	
7	VIEW	VW_DTP_08E3186C	502	23594	
8	SORT UNIQUE		502	23594	
9	TABLE ACCESS FULL	PSP_POSTING_RULE	502	23594	
10	MERGE JOIN CARTESIAN		155M	11G	
11	TABLE ACCESS FULL	PSP_LEDGER_ACCOUNT	1	26	
12	BUFFER SORT		155M	8296M	
13	PARTITION RANGE ALL		155M	8296M	
14	TABLE ACCESS FULL	PSP_FINANCIAL_TRANSACTION	155M	8296M	
15	SORT JOIN		2670K	364M	869M
16	PARTITION RANGE ITERATOR		2670K	364M	
17	TABLE ACCESS BY LOCAL INDEX ROWID BATCHED	PSP_FINANCIAL_TRANS_STATE	2670K	364M	
18	INDEX RANGE SCAN	FN_STATE_EFF_DATE	8359K		

Peeked Binds (identified by position):

- 4 - :B2 (TIMESTAMP): [Not Printable]
- 5 - :B1 (NUMBER): 6
- 6 - :B2 (TIMESTAMP, Primary=4)
- 7 - :B1 (NUMBER, Primary=5)

Predicate Information (identified by operation id):

- 6 - access("ITEM_2"="LA"."LEDGER_ACCOUNT_CD" AND "ITEM_1"="FT"."TRANSACTION_TYPE_FK")
- 9 - filter(("PR"."TRANSACTION_STATE_FK"='Completed' OR "PR"."TRANSACTION_STATE_FK"='Executed' OR "PR"."TRANSACTION_STATE_FK"='Voided'))
- 11 - filter(("LA"."REPORTING_FREQUENCY"='Daily' AND INTERNAL_FUNCTION("LA"."LEDGER_ACCOUNT_TYPE")))
- 14 - filter("FT"."SKU" IS NOT NULL)
- 15 - access("ITEM_1"="FTS"."FINANCIAL_TRANSACTION_FK" AND "ITEM_2"="FTS"."TRANSACTION_STATE_FK")

```

filter(("ITEM_2"="FTS"."TRANSACTION_STATE_FK" AND "ITEM_1"="FTS"."FINANCIAL_TRANSACTION_FK"))
17 - filter((INTERNAL_FUNCTION("FTS"."TRANSACTION_STATE_FK") AND "FTS"."GEMS_UPLOAD_BATCH_FK" IS NULL AND
"FTS"."TRANSACTION_STATE_EFF_DATE">=TRUNC(:B2)-:B1))
18 - access("FTS"."SYS_NC00016$" >= TRUNC(:B2)-:B1)

```

Note

- SQL profile SYS_SQLPR0F_016bfbb5b90d0000 used for this statement

Opportunity #3

BaseATFExtractFileProcess - gjbcjrs2nqth4 . Execution frequency Daily. Physical reads ~1 - 8.6 M

```

/* dynamic native SQL query */ SELECT PC.SOURCE_COMPANY_ID AS PSID,
PE.SOURCE_EMPLOYEE_ID AS EMPLOYEE_ID,
PCL.LAW_FK AS LAW_ID,
MD.VALUE AS VALUE
FROM PSP_TAX_TABLE_MISC_DATA MD
JOIN PSP_EMPLOYEE_TAX PET ON MD.EMPLOYEE_TAX_FK = PET.EMPLOYEE_TAX_SEQ
JOIN PSP_COMPANY_LAW PCL ON PET.COMPANY_LAW_FK = PCL.COMPANY_LAW_SEQ
JOIN PSP_EMPLOYEE PE ON PET.EMPLOYEE_FK = PE.EMPLOYEE_SEQ
JOIN PSP_COMPANY PC ON PE.COMPANY_FK = PC.COMPANY_SEQ
JOIN PSP_COMPANY_SERVICE PCS ON PC.COMPANY_SEQ = PCS.COMPANY_FK
JOIN PSP_TAX_COMPANY_SERVICE_INFO PTCSI
ON PCS.COMPANY_SERVICE_SEQ = PTCSI.TAX_COMPANY_SERVICE_INFO_SEQ
WHERE MD.VALUE IS NOT NULL
AND ((PCS.STATUS_CD = 'ActiveCurrent' AND PCS.SERVICE_START_DATE IS NOT NULL)
OR (PCS.STATUS_CD IN ('Cancelled',
'Terminated') AND PTCSI.LAST_QUARTER_TO_FILE <> 0))
AND MD.MODIFIED_DATE >= :1
order by PC.SOURCE_COMPANY_ID, PE.SOURCE_EMPLOYEE_ID

```

Plan hash value: 2815818285

Id	Operation	Name	Rows	Bytes	TempSpc	Cost
0	SELECT STATEMENT					
491K(100)						
1	SORT ORDER BY		1	432		
491K (1)	00:00:20					
2	NESTED LOOPS		1	432		
491K (1)	00:00:20					
3	NESTED LOOPS		1	391		
491K (1)	00:00:20					
4	NESTED LOOPS		29463	9811K		
377K (1)	00:00:15					
5	NESTED LOOPS		53326	13M		
181K (1)	00:00:08					
6	NESTED LOOPS		6137	1078K		
134K (1)	00:00:06					
* 7	HASH JOIN		6024	782K	7248K	
110K (1)	00:00:05					
8	TABLE ACCESS FULL	PSP_TAX_COMPANY_SERVICE_INFO	140K	5606K		
522 (1)	00:00:01					
9	TABLE ACCESS FULL	PSP_COMPANY_SERVICE	9534K	836M		
62258 (1)	00:00:03					
10	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY	1	47		
4 (0)	00:00:01					
* 11	INDEX RANGE SCAN	SYS_C0020424	1			
2 (0)	00:00:01					
12	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE	9	711		
18 (0)	00:00:01					
* 13	INDEX RANGE SCAN	PSP_EMPLOYEE_FK2	28			
3 (0)	00:00:01					
* 14	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE_TAX	1	82		

```

5  (0)| 00:00:01 |
|* 15 |          INDEX RANGE SCAN                | PSP_EMPLOYEE_TAX_FK2          | 9 |      |      |
3  (0)| 00:00:01 |
|* 16 |          TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_TABLE_MISC_DATA      | 1 | 50 |      |
5  (0)| 00:00:01 |
|* 17 |          INDEX RANGE SCAN                | PSP_TAX_TABLE_MISC_DATA_FK1  | 3 |      |      |
3  (0)| 00:00:01 |
| 18 |          TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_LAW              | 1 | 41 |      |
4  (0)| 00:00:01 |
|* 19 |          INDEX RANGE SCAN                | SYS_C0020562                 | 1 |      |      |
2  (0)| 00:00:01 |

```

Peeked Binds (identified by position):

1 - :1 (TIMESTAMP): [Not Printable]

Predicate Information (identified by operation id):

```

7 - access("PCS"."COMPANY_SERVICE_SEQ"="PTCSI"."TAX_COMPANY_SERVICE_INFO_SEQ")
    filter(((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
              (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSI"."LAST_QUARTER_TO_FILE"<>0)))
11 - access("PC"."COMPANY_SEQ"="PCS"."COMPANY_FK")
13 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
14 - filter("PET"."COMPANY_LAW_FK" IS NOT NULL)
15 - access("PET"."EMPLOYEE_FK"="PE"."EMPLOYEE_SEQ")
16 - filter(("MD"."VALUE" IS NOT NULL AND "MD"."MODIFIED_DATE">=:1))
17 - access("MD"."EMPLOYEE_TAX_FK"="PET"."EMPLOYEE_TAX_SEQ")
19 - access("PET"."COMPANY_LAW_FK"="PCL"."COMPANY_LAW_SEQ")

```

Note

- statistics feedback used for this statement
- this is an adaptive plan

70 rows selected.

Opportunity #4

WorkersCompGatewayManager - 25z1j1qcud1p7 . Execution frequency hourly. Physical reads ~1.1 M

```

/* criteria query */ select this_.COMPANY_EVENT_SEQ      as COMPANY1_32_0_,
                          this_.VERSION                as VERSION32_0_,
                          this_.CREATOR_ID              as CREATOR3_32_0_,
                          this_.CREATED_DATE            as CREATED4_32_0_,
                          this_.MODIFIER_ID             as MODIFIER5_32_0_,
                          this_.MODIFIED_DATE           as MODIFIED6_32_0_,
                          this_.REALM_ID                as REALM7_32_0_,
                          this_.EVENT_TIME_STAMP        as EVENT8_32_0_,
                          this_.STATUS_EFFECTIVE_DATE   as STATUS9_32_0_,
                          this_.STATUS_CD              as STATUS10_32_0_,
                          this_.EVENT_TYPE_CD           as EVENT11_32_0_,
                          this_.EVENT_TOKEN             as EVENT12_32_0_,
                          this_.SOURCE_ID              as SOURCE13_32_0_,
                          this_.NOTE_LAST_UPDATED_DATE  as NOTE14_32_0_,
                          this_.COMPANY_FK              as COMPANY15_32_0_
from PSP_COMPANY_EVENT this_
where (((this_.EVENT_TYPE_CD = :1 or this_.EVENT_TYPE_CD=:2) and this_.STATUS_CD = :3) and
       this_.EVENT_TIME_STAMP between :4 and :5)

```

Plan hash value: 847244176

Id	Operation	Name	Rows	Bytes	Cost (%CPU)	Time
0	SELECT STATEMENT				5 (100)	
* 1	FILTER					
* 2	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY_EVENT	1	189	5 (0)	00:00:01
* 3	INDEX RANGE SCAN	PSP_COMPANY_EVENT_I2	1		4 (0)	00:00:01

Peeked Binds (identified by position):

- ```

1 - :1 (VARCHAR2(30), CSID=873): 'EINChanged'
2 - :2 (VARCHAR2(30), CSID=873): 'NewPSIDCreatedForExistingCustomer'
3 - :3 (VARCHAR2(30), CSID=873): 'Active'
4 - :4 (TIMESTAMP): [Not Printable]
5 - :5 (TIMESTAMP): [Not Printable]

```

Predicate Information (identified by operation id):

- ```

1 - filter(:5>=:4)
2 - filter((INTERNAL_FUNCTION("THIS_". "EVENT_TYPE_CD") AND "THIS_". "STATUS_CD"=:3))
3 - access("THIS_". "EVENT_TIME_STAMP">=:4 AND "THIS_". "EVENT_TIME_STAMP"<=:5)

```

42 rows selected.

Opportunity #5

SQL ordered by Reads (24-Sep-21 18:30 - 24-Sep-21 19:00)

Physical Reads	Executions	Reads per Exec	% Total	Elapsed Time (s)	% CPU	%IO	SQL Id	SQL Module	SQL Text
53,553,913	0		63.41	381.61	46.92	54.18	6qk524uazdf0c	GemsAccountsReceivableProcess or:	UPDATE PSP_FINANCIAL_TRANS_STA...
4,718,576	1	4,718,576.00	5.59	100.30	17.11	25.04	fud06g9jzyyh2	BaseATFExtractFileProcess:146	/* Select companyDailyLiabilit...
4,140,680	1	4,140,680.00	4.90	376.26	13.15	79.19	5nmd8a96rcf9j	BaseATFExtractFileProcess:146	/* Select ptp.Law , ptp.Amount...
2,252,812	1	2,252,812.00	2.67	231.97	8.20	93.73	5ttnr2d161kga	BaseATFExtractFileProcess:146	/* select co, serviceInfo.Stat...

1,286,568	1	1,286,568.00	1.52	159.00	5.51	87.57	a1zfnguds0wgx	BaseATFExtractFileProcess:146	/* select comp.SourceCompanyId...
904,345	1	904,345.00	1.07	538.02	16.38	91.20	2nd0g8cwbz6zg	DailyBatchJobsProcessor:776	BEGIN prc_offload_update_payro...
878,862	1	878,862.00	1.04	509.02	14.20	93.51	1w9qy5w5y3cxf	DailyBatchJobsProcessor:776	UPDATE PSP_PAYROLL_RUN PR SET ...

Opportunity #6

Opportunity #7

BaseATFExtractFileProcess - fud06g9jzyyh2

```

/* Select companyDailyLiability, companyDailyLiability.Company.SourceCompanyId, companyDailyLiability.
Company.SourceSystemCd
from com.intuit.sbd.payroll.psp.domain.CompanyDailyLiability as companyDailyLiability
where companyDailyLiability.LiabilityDate <= :endDate and companyDailyLiability.ModifiedDate >= :lastRunDate
and companyDailyLiability.Company.IsDgDisassociated=0
order by companyDailyLiability.Company, companyDailyLiability.LiabilityDate, companyDailyLiability.Law */
select companydai0_.COMPANY_DAILY_LIABILITY_SEQ as col_0_0_,
       company1_.SOURCE_COMPANY_ID          as col_1_0_,
       company1_.SOURCE_SYSTEM_CD           as col_2_0_,
       companydai0_.COMPANY_DAILY_LIABILITY_SEQ as COMPANY1_31_,
       companydai0_.VERSION                  as VERSION31_,
       companydai0_.CREATOR_ID               as CREATOR3_31_,
       companydai0_.CREATED_DATE             as CREATED4_31_,
       companydai0_.MODIFIER_ID              as MODIFIER5_3_1_,
       companydai0_.MODIFIED_DATE            as MODIFIED6_31_,
       companydai0_.REALM_ID                 as REALM7_31_,
       companydai0_.LIABILITY_DATE           as LIABILITY8_31_,
       companydai0_.TAXABLE_WAGES            as TAXABLE9_31_,
       companydai0_.TOTAL_WAGES              as TOTAL10_31_,
       companydai0_.TOTAL_TIPS_AMOUNT        as TOTAL11_31_,
       companydai0_.TAX_AMOUNT               as TAX12_31_,
       companydai0_.LAW_FK                   as LAW13_31_,
       companydai0_.COMPANY_FK               as COMPANY14_31_
from PSP_COMPANY_DAILY_LIABILITY companydai0_,
     PSP_COMPANY company1_
where companydai0_.COMPANY_FK = company1_.COMPANY_SEQ
and companydai0_.LIABILITY_DATE <= : 1
and companydai0_.MODIFIED_DATE >=: 2
and company1_.IS_DG_DISASSOCIATED = 0
order by
       companydai0_.COMPANY_FK,
       companydai0_.LIABILITY_DATE,
       companydai0_.LAW_FK

```

Plan hash value: 2440059379

Id	Operation	Name	Rows	Bytes	Cost (%CPU)	Time	
0	SELECT STATEMENT				1282K(100)		
1	SORT ORDER BY		8	1840	1282K (1)	00:00:51	
2	NESTED LOOPS		8	1840	1282K (1)	00:00:51	
3	NESTED LOOPS		8	1840	1282K (1)	00:00:51	
* 4	TABLE ACCESS FULL	PSP_COMPANY_DAILY_LIABILITY	8	1400	1282K (1)	00:00:51	
* 5	INDEX RANGE SCAN	SYS_C0020424	1		2 (0)	00:00:01	
* 6	TABLE ACCESS BY INDEX ROWID	PSP_COMPANY	1	55	4 (0)	00:00:01	

Predicate Information (identified by operation id):

- ```

4 - filter(("COMPANYDAI0"."MODIFIED_DATE">=:2 AND "COMPANYDAI0"."LIABILITY_DATE"<=:1))
5 - access("COMPANYDAI0"."COMPANY_FK"="COMPANY1"."COMPANY_SEQ")
6 - filter(NVL("COMPANY1"."IS_DG_DISASSOCIATED",0)=0)

```

Opportunity #8

BaseATFExtractFileProcess - 5nmd8a96rcf9j

```
/* Select ptp.Law
, ptp.Amount
, ptp.Company.SourceCompanyId, ptp.Company.SourceSystemCd
, ptp.QuarterEndDate, ptp.PaymentDate, ptp.MoneyMovementTransaction.Id
from com.intuit.sbd.payroll.psp.domain.ATFPaymentsToProcess as ptp
where ptp.QuarterEndDate >= :startDate and ptp.QuarterEndDate <= :endDate
and exists (Select 1 from com.intuit.sbd.payroll.psp.domain.ATFPaymentsToProcess as updatedCos where updatedCos.
Company = ptp.Company and updatedCos.ModifiedDate >= :lastRunDate)
and ptp.Company.IsDgDisassociated=0
order by ptp.Company.SourceCompanyId, ptp.PaymentDate, ptp.Law.PaymentTemplate, ptp.MoneyMovementTransaction.
Id, ptp.Law */
select atfpayment0_.LAW_FK as col_0_0_,
atfpayment0_.AMOUNT as col_1_0_,
company2_.SOURCE_COMPANY_ID as col_2_0_,
company2_.SOURCE_SYSTEM_CD as col_3_0_,
atfpayment0_.QUARTER_END_DATE as col_4_0_,
atfpayment0_.PAYMENT_DATE as col_5_0_,
atfpayment0_.MONEY_MOVEMENT_TRANSACTION_FK as col_6_0_,
law1_.LAW_ID as LAW1_242_,
law1_.VERSION as VERSION242_,
law1_.REALM_ID as REALM3_242_,
law1_.DESCRIPTION as DESCRIPT4_242_,
law1_.LAW_CATEGORY_CODE as LAW5_242_,
law1_.LAW_ABBREV as LAW6_242_,
law1_.IS_EMPLOYER_TAX as IS7_242_,
law1_.LAW_TYPE_CD as LAW8_242_,
law1_.REQUIRES_MONTH_COUNTS as REQUIRES9_242_,
law1_.PAYMENT_TEMPLATE_FK as PAYMENT10_242_
from PSP_ATFPAYMENTS_TO_PROCESS atfpayment0_
inner join PSP_LAW law1_ on atfpayment0_.LAW_FK = law1_.LAW_ID,
PSP_COMPANY company2_,
PSP_LAW law7_
where atfpayment0_.COMPANY_FK = company2_.COMPANY_SEQ
and atfpayment0_.LAW_FK = law7_.LAW_ID
and atfpayment0_.QUARTER_END_DATE >= : 1
and atfpayment0_.QUARTER_END_DATE <=: 2
and (
exists (
select
1
from
PSP_ATFPAYMENTS_TO_PROCESS atfpayment4_
where
atfpayment4_.COMPANY_FK = atfpayment0_.COMPANY_FK
and atfpayment4_.MODIFIED_DATE >=: 3
)
)
and company2_.IS_DG_DISASSOCIATED = 0
order by
comp any2_.SOURCE_COMPANY_ID,
atfpayment0_.PAYMENT_DATE,
law7_.PAYMENT_TEMPLATE_FK,
atfpayment0_.MONEY_MOVEMENT_TRANSACTION_FK,
atfpayment0_.LAW_FK
```

Plan hash value: 2548189497

| ----- |                  |      |      |       |         |          |
|-------|------------------|------|------|-------|---------|----------|
| Id    | Operation        | Name | Rows | Bytes | TempSpc | Cost (%) |
| ----- |                  |      |      |       |         |          |
| 0     | SELECT STATEMENT |      |      |       |         | 890K     |
| (100) |                  |      |      |       |         |          |



|      |                                     |                            |      |       |       |      |
|------|-------------------------------------|----------------------------|------|-------|-------|------|
| 1    | SORT ORDER BY                       |                            | 8247 | 2504K | 2648K | 890K |
| (1)  | 00:00:35                            |                            |      |       |       |      |
| * 2  | FILTER                              |                            |      |       |       |      |
| 3    | NESTED LOOPS                        |                            | 8247 | 2504K |       | 890K |
| (1)  | 00:00:35                            |                            |      |       |       |      |
| * 4  | HASH JOIN                           |                            | 8093 | 2023K |       | 857K |
| (1)  | 00:00:34                            |                            |      |       |       |      |
| 5    | VIEW                                | index\$_join\$_005         | 238  | 4760  |       | 2    |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| * 6  | HASH JOIN                           |                            |      |       |       |      |
| 7    | INDEX FAST FULL SCAN                | PSP_LAW_FK1                | 238  | 4760  |       | 1    |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| 8    | INDEX FAST FULL SCAN                | SYS_C0020547               | 238  | 4760  |       | 1    |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| * 9  | HASH JOIN                           |                            | 8093 | 1865K |       | 857K |
| (1)  | 00:00:34                            |                            |      |       |       |      |
| 10   | TABLE ACCESS FULL                   | PSP_LAW                    | 238  | 19992 |       | 4    |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| * 11 | HASH JOIN                           |                            | 8093 | 1201K |       | 857K |
| (1)  | 00:00:34                            |                            |      |       |       |      |
| 12   | SORT UNIQUE                         |                            | 59   | 2832  |       | 848K |
| (1)  | 00:00:34                            |                            |      |       |       |      |
| * 13 | TABLE ACCESS FULL                   | PSP_ATFPAYMENTS_TO_PROCESS | 59   | 2832  |       | 848K |
| (1)  | 00:00:34                            |                            |      |       |       |      |
| * 14 | TABLE ACCESS FULL                   | PSP_ATFPAYMENTS_TO_PROCESS | 138  | 14352 |       | 309  |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| * 15 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                | 1    | 55    |       | 4    |
| (0)  | 00:00:01                            |                            |      |       |       |      |
| * 16 | INDEX RANGE SCAN                    | SYS_C0020424               | 1    |       |       | 2    |
| (0)  | 00:00:01                            |                            |      |       |       |      |

Predicate Information (identified by operation id):

```

2 - filter(:2>=:1)
4 - access("ATFPAYMENT0"."LAW_FK"="LAW7"."LAW_ID")
6 - access(ROWID=ROWID)
9 - access("ATFPAYMENT0"."LAW_FK"="LAW1"."LAW_ID")
11 - access("ATFPAYMENT4"."COMPANY_FK"="ATFPAYMENT0"."COMPANY_FK")
13 - filter("ATFPAYMENT4"."MODIFIED_DATE">=:3)
14 - filter(("ATFPAYMENT0"."QUARTER_END_DATE">=:1 AND "ATFPAYMENT0"."QUARTER_END_DATE"<=:2))
15 - filter(NVL("COMPANY2"."IS_DG_DISASSOCIATED",0)=0)
16 - access("ATFPAYMENT0"."COMPANY_FK"="COMPANY2"."COMPANY_SEQ")

```

Note

- this is an adaptive plan

## Opportunity #9

### BaseATFExtractFileProcess - 5ttnr2d161kga

```

/* select co, serviceInfo.StatusCd, serviceInfo.LastQuarterToFile,
serviceInfo.FinalAnnualReturns, serviceInfo.FileAnnualReturns, serviceInfo.ServiceStartDate, serviceInfo.
LastPayrollDate
from com.intuit.sbd.payroll.psp.domain.TaxCompanyServiceInfo serviceInfo
join serviceInfo.Company co
join fetch co.LegalAddress address
left join fetch co.OnHoldReasonSet reason
where

```

```

serviceInfo.ServiceStartDate is not null and serviceInfo.StatusCd in ('ActiveCurrent', 'Cancelled',
'Terminated')
and (co.ModifiedDate >= :startDate or address.ModifiedDate >= :startDate
or serviceInfo.ModifiedDate >= :startDate or reason.ModifiedDate >= :startDate)
and co.IsDgDisassociated=0
order by co.SourceCompanyId
*/
select company1_.COMPANY_SEQ as col_0_0_,
taxcompany0_1_.STATUS_CD as col_1_0_,
taxcompany0_.LAST_QUA RTER_TO_FILE as col_2_0_,
taxcompany0_.FINAL_ANNUAL_RETURNS as col_3_0_,
taxcompany0_.FILE_ANNUAL_RETURNS as col_4_0_,
taxcompany0_1_.SERVICE_START_DATE as col_5_0_,
taxcompany0_.LAST_PAYROLL_DATE as col_6_0_,
address2_.ADDRESS_SEQ as ADDRESS1_190_2_,
onholdreas3_.ON_HOLD_REASON_SEQ as ON1_122_3_,
company1_.COMPANY_SEQ as COMPANY1_194_0_,
address2_.ADDRESS_SEQ as ADDRESS1_190_1_,
onholdreas3_.ON_HOLD_REASON_SEQ as ON1_122_2_,
company1_.VERSION as VERSION194_0_,
company1_.CREATOR_ID as CREATOR3_194_0_,
company1_.CREATED_DATE as CREATED4_194_0_,
company1_.MODIFIER_ID as MODIFIER5_194_0_,
company1_.MODIFIED_DATE as MODIFIED6_194_0_,
company1_.REALM_ID as REALM7_194_0_,
company1_.PRIVATE_KEY_ENC as PRIVATE8_194_0_,
company1_.PHONE as PHONE194_0_,
company1_.DBA_NAME as DBA10_194_0_,
company1_.LEGAL_NAME as LEGAL11_194_0_,
company1_.SOURCE_COMPANY_ID as SOURCE12_194_0_,
company1_.DEBUG_LOGGING as DEBUG13_194_0_,
company1_.NOTIFICATION_EMAIL as NOTIFIC14_194_0_,
company 1_.NEXT_PAYROLL_TRANSACTION_ID as NEXT15_194_0_,
company1_.NEXT_PAYCHECK_ID as NEXT16_194_0_,
company1_.NEXT_EMPLOYEE_ID as NEXT17_194_0_,
company1_.NEXT_PAYROLL_ITEM_ID as NEXT18_194_0_,
company1_.ACCOUNT_LOCKED_UNTIL as ACCOUNT19_194_0_,
company1_.NBR_OF_FAILED_LOGIN_ATTEMPTS as NBR20_194_0_,
company1_.CURRENT_TOKEN as CURRENT21_194_0_,
company1_.SOURCE_SYSTEM_CD as SOURCE22_194_0_,
company1_.TAX_EXEMPT_EXPIRATION_DATE as TAX23_194_0_,
company1_.NAME_CONTROL as NAME24_194_0_,
company1_.IS_FLAGGED_FOR_FRAUD as IS25_194_0_,
company1_.SIGN_UP_DATE as SIGN26_194_0_,
company1_.PUBLIC_KEY as PUBLIC27_194_0_,
company1_.NBR_FAILED_AUTHENTICATIONS as NBR28_194_0_,
company1_.CLOUD_CURRENT_TOKEN as CLOUD29_194_0_,
company1_.I_A_M_REALM_ID as I30_194_0_,
company1_.TAX_EXEMPT_STATUS as TAX31_194_0_,
company1_.PRICE_TYPE as PRICE32_194_0_,
company1_.FED_TAX_ID_ENC as FED33_194_0_,
company1_.D_D_PUBLISH_FLAG as D34_194_0_,
company1_.O_I_I_FLAG as O35_194_0_,
company1_.IS_DG_DIS ASSOCIATED as IS36_194_0_,
company1_.PUBLISH_STATUS as PUBLISH37_194_0_,
company1_.COMPLIANCE_ADDRESS_FK as COMPLIANCE38_194_0_,
company1_.MAILING_ADDRESS_FK as MAILING39_194_0_,
company1_.LEGAL_ADDRESS_FK as LEGAL40_194_0_,
company1_.PAYROLL_FREQUENCY_FK as PAYROLL41_194_0_,
company1_.OFFLOAD_GROUP_FK as OFFLOAD42_194_0_,
company1_.FUNDING_MODEL_FK as FUNDING43_194_0_,
address2_.VERSION as VERSION190_1_,
address2_.CREATOR_ID as CREATOR3_190_1_,
address2_.CREATED_DATE as CREATED4_190_1_,
address2_.MODIFIER_ID as MODIFIER5_190_1_,
address2_.MODIFIED_DATE as MODIFIED6_190_1_,
address2_.REALM_ID as REALM7_190_1_,
address2_.ADDRESS_LINE3 as ADDRESS8_190_1_,
address2_.ADDRESS_LINE2 as ADDRESS9_190_1_,
address2_.ADDRESS_LINE1 as ADDRESS10_190_1_,
address2_.CITY as CITY190_1_,

```

```

address2_.COUNTRY as COUNTRY190_1_,
address2_.STATE as STATE190_1_,
address2_.ZIP_CODE as ZIP14_190_1_,
address2_.ZIP_CODE_EXTENSION as ZIP15_190_1_,
onholdreas3_.VERSION as VERSION122_2_,
onholdreas3_.CREATOR_ID as CREATOR3_122_2_,
onholdreas3_.CREATED_DATE as CREATED4_122_2_,
onholdreas3_.MODIFIER_ID as MODIFIER5_122_2_,
onholdreas3_.MODIFIED_DATE as MODIFIED6_122_2_,
onholdreas3_.REALM_ID as REALM7_122_2_,
onholdreas3_.EFFECTIVE_DATE as EFFECTIVE8_122_2_,
onholdreas3_.EXPIRATION_DATE as EXPIRATION9_122_2_,
onholdreas3_.ON_HOLD_REASON_CD as ON10_122_2_,
onholdreas3_.COMPANY_FK as COMPANY11_122_2_,
onholdreas3_.COMPANY_FK as COMPANY11_194_0_,
onholdreas3_.ON_HOLD_REASON_SEQ as ON1_0_
from PSP_TAX_COMPANY_SERVICE_INFO taxcompany0_
 inner join PSP_COMPANY_SERVICE taxcompany0_1_
 on taxcompany0_.TAX_COMPANY_SERVICE_INFO_SEQ = taxcompany0_1_.COMPANY_SERVICE_SEQ
 inner join PSP_COMPANY company1_ on taxcompany0_1_.COMPANY_FK = company1_.COMPANY_SEQ
 inner join PSP_ADDRESS address2_ on company1_.LEGAL_ADDRESS_FK = address2_.ADDRESS_SEQ
 left outer join PSP_ON_HOLD_REASON onholdreas3_ on company1_.COMPANY_SEQ = onholdreas3_.COMPANY_FK
where (
 taxcompany0_1_.SERVICE_START_DATE is not null
)
and (
 taxcompany0_1_.STATUS_CD in (
 'ActiveCurrent', 'Cancelled', 'Terminated'
)
)
and (
 company1_.MODIFIED_DATE >= : 1
 or address2_.MODIFIED_DATE >= : 2
 or taxcompany0_1_.MODIFIED_DATE >=: 3
 or onholdreas3_.MODIFIED_DATE >=: 4
)
and company1_.IS_DG_DISASSOCIATED = 0
order by company1_.SOURCE_COMPANY_ID

```

Plan hash value: 3616741784

| Id       |  | Operation                           | Name                         | Rows | Bytes | Cost (%CPU) |
|----------|--|-------------------------------------|------------------------------|------|-------|-------------|
| Time     |  |                                     |                              |      |       |             |
| 0        |  | SELECT STATEMENT                    |                              |      |       | 701K(100)   |
| 1        |  | SORT ORDER BY                       |                              | 11   | 185K  | 701K (1)    |
| 00:00:28 |  |                                     |                              |      |       |             |
| 2        |  | VIEW                                | VW_ORE_AA72F09C              | 11   | 185K  | 701K (1)    |
| 00:00:28 |  |                                     |                              |      |       |             |
| 3        |  | UNION-ALL                           |                              |      |       |             |
| 4        |  | NESTED LOOPS OUTER                  |                              | 8    | 9408  | 195K (1)    |
| 00:00:08 |  |                                     |                              |      |       |             |
| 5        |  | NESTED LOOPS                        |                              | 4    | 4020  | 195K (1)    |
| 00:00:08 |  |                                     |                              |      |       |             |
| 6        |  | NESTED LOOPS                        |                              | 4    | 3536  | 195K (1)    |
| 00:00:08 |  |                                     |                              |      |       |             |
| 7        |  | NESTED LOOPS                        |                              | 4    | 3324  | 195K (1)    |
| 00:00:08 |  |                                     |                              |      |       |             |
| 8        |  | TABLE ACCESS FULL                   | PSP_COMPANY                  | 4    | 2956  | 195K (1)    |
| 00:00:08 |  |                                     |                              |      |       |             |
| 9        |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE          | 1    | 92    | 4 (0)       |
| 00:00:01 |  |                                     |                              |      |       |             |
| 10       |  | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1      | 1    |       | 2 (0)       |
| 00:00:01 |  |                                     |                              |      |       |             |
| 11       |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_COMPANY_SERVICE_INFO | 1    | 53    | 4 (0)       |

|          |    |                                     |                              |   |      |       |     |
|----------|----|-------------------------------------|------------------------------|---|------|-------|-----|
| 00:00:01 | 12 | INDEX RANGE SCAN                    | SYS_C0020664                 | 1 |      | 2     | (0) |
| 00:00:01 | 13 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ADDRESS                  | 1 | 121  | 5     | (0) |
| 00:00:01 | 14 | INDEX RANGE SCAN                    | SYS_C0020476                 | 1 |      | 3     | (0) |
| 00:00:01 | 15 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ON_HOLD_REASON           | 2 | 342  | 5     | (0) |
| 00:00:01 | 16 | INDEX RANGE SCAN                    | PSP_ON_HOLD_REASON_FK1       | 2 |      | 2     | (0) |
| 00:00:01 | 17 | NESTED LOOPS OUTER                  |                              | 1 | 1176 | 442K  | (1) |
| 00:00:18 | 18 | NESTED LOOPS                        |                              | 1 | 1005 | 442K  | (1) |
| 00:00:18 | 19 | NESTED LOOPS                        |                              | 1 | 952  | 442K  | (1) |
| 00:00:18 | 20 | NESTED LOOPS                        |                              | 1 | 860  | 442K  | (1) |
| 00:00:18 | 21 | TABLE ACCESS FULL                   | PSP_ADDRESS                  | 1 | 121  | 442K  | (1) |
| 00:00:18 | 22 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1 | 739  | 4     | (0) |
| 00:00:01 | 23 | INDEX RANGE SCAN                    | PSP_COMPANY_FK4              | 1 |      | 2     | (0) |
| 00:00:01 | 24 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE          | 1 | 92   | 4     | (0) |
| 00:00:01 | 25 | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1      | 1 |      | 2     | (0) |
| 00:00:01 | 26 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_COMPANY_SERVICE_INFO | 1 | 53   | 4     | (0) |
| 00:00:01 | 27 | INDEX RANGE SCAN                    | SYS_C0020664                 | 1 |      | 2     | (0) |
| 00:00:01 | 28 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ON_HOLD_REASON           | 1 | 171  | 5     | (0) |
| 00:00:01 | 29 | INDEX RANGE SCAN                    | PSP_ON_HOLD_REASON_FK1       | 2 |      | 2     | (0) |
| 00:00:01 | 30 | NESTED LOOPS OUTER                  |                              | 1 | 1187 | 60057 | (1) |
| 00:00:03 | 31 | NESTED LOOPS                        |                              | 1 | 1016 | 60052 | (1) |
| 00:00:03 | 32 | NESTED LOOPS                        |                              | 1 | 895  | 60047 | (1) |
| 00:00:03 | 33 | NESTED LOOPS                        |                              | 1 | 156  | 60043 | (1) |
| 00:00:03 | 34 | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE          | 1 | 103  | 60039 | (1) |
| 00:00:03 | 35 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_COMPANY_SERVICE_INFO | 1 | 53   | 4     | (0) |
| 00:00:01 | 36 | INDEX RANGE SCAN                    | SYS_C0020664                 | 1 |      | 2     | (0) |
| 00:00:01 | 37 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1 | 739  | 4     | (0) |
| 00:00:01 | 38 | INDEX RANGE SCAN                    | SYS_C0020424                 | 1 |      | 2     | (0) |
| 00:00:01 | 39 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ADDRESS                  | 1 | 121  | 5     | (0) |
| 00:00:01 | 40 | INDEX RANGE SCAN                    | SYS_C0020476                 | 1 |      | 3     | (0) |
| 00:00:01 | 41 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ON_HOLD_REASON           | 1 | 171  | 5     | (0) |
| 00:00:01 | 42 | INDEX RANGE SCAN                    | PSP_ON_HOLD_REASON_FK1       | 2 |      | 2     | (0) |
| 00:00:01 | 43 | NESTED LOOPS                        |                              | 1 | 1187 | 2788  | (1) |
| 00:00:01 | 44 | NESTED LOOPS                        |                              | 1 | 1066 | 2783  | (1) |
| 00:00:01 | 45 | NESTED LOOPS                        |                              | 1 | 1013 | 2779  | (1) |
| 00:00:01 | 46 | NESTED LOOPS                        |                              | 1 | 910  | 2775  | (1) |
| 00:00:01 |    |                                     |                              |   |      |       |     |

|          |                                     |                              |   |     |      |     |
|----------|-------------------------------------|------------------------------|---|-----|------|-----|
| 47       | TABLE ACCESS FULL                   | PSP_ON_HOLD_REASON           | 1 | 171 | 2771 | (1) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 48       | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1 | 739 | 4    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 49       | INDEX RANGE SCAN                    | SYS_C0020424                 | 1 |     | 2    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 50       | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE          | 1 | 103 | 4    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 51       | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1      | 1 |     | 2    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 52       | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_COMPANY_SERVICE_INFO | 1 | 53  | 4    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 53       | INDEX RANGE SCAN                    | SYS_C0020664                 | 1 |     | 2    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 54       | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ADDRESS                  | 1 | 121 | 5    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |
| 55       | INDEX RANGE SCAN                    | SYS_C0020476                 | 1 |     | 3    | (0) |
| 00:00:01 |                                     |                              |   |     |      |     |

Note

- this is an adaptive plan

## Opportunity #10

### BaseATFExtractFileProcess - a1zfnguds0wgx

```

/* select comp.SourceCompanyId, ssla.SourceLawCode, capt.AgencyTaxpayerIdEnc, complaw.FilingStatus, complaw.
ExemptionStatus
 from com.intuit.sbd.payroll.psp.domain.TaxCompanyServiceInfo serviceInfo,
 com.intuit.sbd.payroll.psp.domain.SourceSystemLawAssoc ssla
 join serviceInfo.Company comp
 join comp.CompanyAgencySet compagency
 join compagency.CompanyLawSet complaw
 join compagency.CompanyAgencyPaymentTemplateSet capt
 join complaw.QbdtPayrollItemInfoSet qbdtptiteminfo

where
 ssla.Law = complaw.Law
 and ssla.SourceSystem.SourceSystemCd = comp.SourceSystemCd
 and capt.PaymentTemplate = complaw.Law.PaymentTemplate
 and qbdtptiteminfo.IsDeleted=false
 and (serviceInfo.ServiceStartDate is not nul
l and serviceInfo.StatusCd in ('ActiveCurrent', 'Cancelled', 'Terminated')) and complaw.
AdditionalCompanyLaw is null and (complaw.ModifiedDate >= :startDate or capt.ModifiedDate >= :startDate or
serviceInfo.ModifiedDate >= :startDate or qbdtptiteminfo.ModifiedDate >=:startDate) and comp.
IsDgDisassociated=0
order by comp.SourceCompanyId, ssla.SourceLawCode
*/
select company2_.SOURCE_COMPANY_ID as col_0_0_,
 sourcesyst1_.SOURCE_LAW_CODE as col_1_0_,
 companyage5_.AGENCY_TAXPAYER_ID_ENC as col_2_0_,
 companylaw4_.FILING_STATUS as col_3_0_,
 companylaw4_.EXEMPTION_STATUS as col_4_0_
from PSP_TAX_COMPANY_SERVICE_INFO taxcompany0_
 inner join PSP_COMPANY_SERVICE taxcompany0_1_
 on taxcompany0_.TAX_COMPANY_SERVICE_INFO_SEQ = taxcompany0_1_.COMPANY_SERVICE_SEQ
 inner join PSP_COMPANY company2_ on taxcompany0_1_.COMPANY_FK = company2_.COMPANY_SEQ
 inner join PSP_COMPANY_AGENCY companyage3_ on company2_.COMPANY_SEQ = companyage3_.COMPANY_FK
 inner join PSP_COMPANY_LAW companylaw4_
on companyage3_.COMPANY_AGENCY_SEQ = companylaw4_.COMPANY_AGENCY_FK
 inner join PSP_QBDT_PAYROLL_ITEM_INFO qbdtpayrol6_ on companylaw4_.COMPANY_LAW_SEQ = qbdtpayrol6_.
COMPANY_LAW_FK
 inner join PSP_COMPANYAGENCY_PMTTEMPLATE companyage5_ on companyage3_.COMPANY_AGENCY_SEQ = companyage5_.
COMPANY_AGENCY_FK,
 PSP_SOURCE_SYSTEM_LAW_ASSOC sourcesyst1_,
 PSP_LAW law7_
where

```

```

 companylaw4_.LAW_FK = law7_.LAW_ID
and sourcesyst1_.LAW_FK = companylaw4_.LAW_FK
and sourcesyst1_.SOURCE_SYSTEM_FK = company2_.SOURCE_SYSTEM_CD
and companyage5_.PAYMENT_TEMPLATE_FK = law7_.PAYMENT_TEMPLATE_FK
and qbdtpayrol6_.IS_DELETED = 0
and (
 taxcompany0_1_.SERVICE_START_DATE is not null
)
and (
 taxcompany0_1_.STATUS_CD in (
 'ActiveCurrent'
 , 'Cancelled'
 , 'Terminated'
)
)
and (
 companylaw4_.ADDITIONAL_COMPANY_LAW_FK is null
)
and (
 companylaw4_.MODIFIED_DATE >=: 1
or companyage5_.MODIFIED_DATE >=: 2
or taxcompany0_1_.MODIFIED_DATE >=: 3
or qbdtpayrol6_.MODIFIED_DATE >=: 4
)
and company2_.IS_DG_DISASSOCIATED = 0
order by
 company 2_.SOURCE_COMPANY_ID,
 sourcesyst1_.SOURCE_LAW_CODE

```

Plan hash value: 2100946717

| Id          | Operation                           | Name                       | Rows | Bytes |
|-------------|-------------------------------------|----------------------------|------|-------|
| Cost (%CPU) | Time                                |                            |      |       |
| 0           | SELECT STATEMENT                    |                            |      |       |
| 434K(100)   |                                     |                            |      |       |
| 1           | SORT ORDER BY                       |                            | 4    | 13160 |
| 434K (1)    | 00:00:17                            |                            |      |       |
| 2           | VIEW                                | VW_ORE_E457A751            | 4    | 13160 |
| 434K (1)    | 00:00:17                            |                            |      |       |
| 3           | UNION-ALL                           |                            |      |       |
| 4           | NESTED LOOPS                        |                            | 1    | 573   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 5           | NESTED LOOPS                        |                            | 1    | 536   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 6           | NESTED LOOPS                        |                            | 1    | 433   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 7           | NESTED LOOPS                        |                            | 1    | 378   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| * 8         | HASH JOIN                           |                            | 1    | 357   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 9           | NESTED LOOPS                        |                            | 2    | 674   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 10          | NESTED LOOPS                        |                            | 2    | 420   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| 11          | NESTED LOOPS                        |                            | 2    | 272   |
| 282K (1)    | 00:00:12                            |                            |      |       |
| * 12        | TABLE ACCESS FULL                   | PSP_QBDT_PAYROLL_ITEM_INFO | 2    | 54    |
| 282K (1)    | 00:00:12                            |                            |      |       |
| * 13        | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_LAW            | 1    | 109   |
| 4 (0)       | 00:00:01                            |                            |      |       |
| * 14        | INDEX RANGE SCAN                    | SYS_C0020562               | 1    |       |
| 2 (0)       | 00:00:01                            |                            |      |       |
| 15          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY         | 1    | 74    |
| 3 (0)       | 00:00:01                            |                            |      |       |
| * 16        | INDEX RANGE SCAN                    | SYS_C0020551               | 1    |       |

|       |     |          |                                     |                                |     |      |
|-------|-----|----------|-------------------------------------|--------------------------------|-----|------|
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 17  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANYAGENCY_PMTTEMPLATE  | 1   | 127  |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 18  |          | INDEX RANGE SCAN                    | PSP_COMPANY_AGENCY_PAYMENT_FK1 | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 19  |          | VIEW                                | index\$_join\$_070             | 238 | 4760 |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 20  |          | HASH JOIN                           |                                |     |      |
|       |     |          |                                     |                                |     |      |
|       | 21  |          | INDEX FAST FULL SCAN                | PSP_LAW_FK1                    | 238 | 4760 |
| 1     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 22  |          | INDEX FAST FULL SCAN                | SYS_C0020547                   | 238 | 4760 |
| 1     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 23  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_SOURCE_SYSTEM_LAW_ASSOC    | 2   | 42   |
| 3     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 24  |          | INDEX RANGE SCAN                    | PSP_SOURCE_SYSTEM_LAW_ASSO_FK1 | 2   |      |
| 1     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 25  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                    | 1   | 55   |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 26  |          | INDEX RANGE SCAN                    | SYS_C0020424                   | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 27  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE            | 1   | 103  |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 28  |          | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1        | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 29  |          | INDEX RANGE SCAN                    | SYS_C0020664                   | 1   | 37   |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 30  |          | NESTED LOOPS                        |                                | 1   | 551  |
| 23054 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 31  |          | NESTED LOOPS                        |                                | 1   | 535  |
| 23051 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 32  |          | NESTED LOOPS                        |                                | 1   | 498  |
| 23049 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 33  |          | NESTED LOOPS                        |                                | 1   | 406  |
| 23045 | (1) | 00:00:01 |                                     |                                |     |      |
| *     | 34  |          | HASH JOIN                           |                                | 1   | 351  |
| 23041 | (1) | 00:00:01 |                                     |                                |     |      |
| *     | 35  |          | HASH JOIN                           |                                | 1   | 330  |
| 23038 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 36  |          | NESTED LOOPS                        |                                | 12  | 3720 |
| 23036 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 37  |          | NESTED LOOPS                        |                                | 3   | 603  |
| 23024 | (1) | 00:00:01 |                                     |                                |     |      |
| *     | 38  |          | TABLE ACCESS FULL                   | PSP_COMPANYAGENCY_PMTTEMPLATE  | 3   | 381  |
| 23015 | (1) | 00:00:01 |                                     |                                |     |      |
|       | 39  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY             | 1   | 74   |
| 3     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 40  |          | INDEX RANGE SCAN                    | SYS_C0020551                   | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 41  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_LAW                | 4   | 436  |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 42  |          | INDEX RANGE SCAN                    | PSP_COMPANY_LAW_FK1            | 4   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 43  |          | VIEW                                | index\$_join\$_052             | 238 | 4760 |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 44  |          | HASH JOIN                           |                                |     |      |
|       |     |          |                                     |                                |     |      |
|       | 45  |          | INDEX FAST FULL SCAN                | PSP_LAW_FK1                    | 238 | 4760 |
| 1     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 46  |          | INDEX FAST FULL SCAN                | SYS_C0020547                   | 238 | 4760 |
| 1     | (0) | 00:00:01 |                                     |                                |     |      |
|       | 47  |          | TABLE ACCESS FULL                   | PSP_SOURCE_SYSTEM_LAW_ASSOC    | 2   | 42   |
| 3     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 48  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                    | 1   | 55   |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 49  |          | INDEX RANGE SCAN                    | SYS_C0020424                   | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 50  |          | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE            | 1   | 92   |
| 4     | (0) | 00:00:01 |                                     |                                |     |      |
| *     | 51  |          | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1        | 1   |      |
| 2     | (0) | 00:00:01 |                                     |                                |     |      |

|           |                                     |                                |  |     |      |
|-----------|-------------------------------------|--------------------------------|--|-----|------|
| * 52      | INDEX RANGE SCAN                    | SYS_C0020664                   |  | 1   | 37   |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 53      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_PAYROLL_ITEM_INFO     |  | 1   | 16   |
| 3 (0)     | 00:00:01                            |                                |  |     |      |
| * 54      | INDEX RANGE SCAN                    | PSP_QBDT_PAYROLL_ITEM_INFO_FK1 |  | 1   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 55      | HASH JOIN                           |                                |  | 2   | 8668 |
| 128K (1)  | 00:00:06                            |                                |  |     |      |
| 56        | VIEW                                | index\$_join\$_061             |  | 238 | 4760 |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 57      | HASH JOIN                           |                                |  |     |      |
|           |                                     |                                |  |     |      |
| 58        | INDEX FAST FULL SCAN                | PSP_LAW_FK1                    |  | 238 | 4760 |
| 1 (0)     | 00:00:01                            |                                |  |     |      |
| 59        | INDEX FAST FULL SCAN                | SYS_C0020547                   |  | 238 | 4760 |
| 1 (0)     | 00:00:01                            |                                |  |     |      |
| 60        | VIEW                                | VW_JF_SET\$18F2F18F            |  | 2   | 8628 |
| 128K (1)  | 00:00:06                            |                                |  |     |      |
| 61        | UNION-ALL                           |                                |  |     |      |
|           |                                     |                                |  |     |      |
| 62        | NESTED LOOPS                        |                                |  | 1   | 542  |
| 60063 (1) | 00:00:03                            |                                |  |     |      |
| 63        | NESTED LOOPS                        |                                |  | 1   | 415  |
| 60059 (1) | 00:00:03                            |                                |  |     |      |
| * 64      | HASH JOIN                           |                                |  | 1   | 399  |
| 60056 (1) | 00:00:03                            |                                |  |     |      |
| * 65      | HASH JOIN                           |                                |  | 1   | 378  |
| 60053 (1) | 00:00:03                            |                                |  |     |      |
| 66        | NESTED LOOPS                        |                                |  | 1   | 341  |
| 60051 (1) | 00:00:03                            |                                |  |     |      |
| 67        | NESTED LOOPS                        |                                |  | 1   | 232  |
| 60047 (1) | 00:00:03                            |                                |  |     |      |
| 68        | NESTED LOOPS                        |                                |  | 1   | 158  |
| 60043 (1) | 00:00:03                            |                                |  |     |      |
| * 69      | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE            |  | 1   | 103  |
| 60039 (1) | 00:00:03                            |                                |  |     |      |
| * 70      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                    |  | 1   | 55   |
| 4 (0)     | 00:00:01                            |                                |  |     |      |
| * 71      | INDEX RANGE SCAN                    | SYS_C0020424                   |  | 1   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| 72        | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY             |  | 1   | 74   |
| 4 (0)     | 00:00:01                            |                                |  |     |      |
| * 73      | INDEX RANGE SCAN                    | PSP_COMPANY_AGENCY_FK2         |  | 4   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 74      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_LAW                |  | 4   | 436  |
| 4 (0)     | 00:00:01                            |                                |  |     |      |
| * 75      | INDEX RANGE SCAN                    | PSP_COMPANY_LAW_FK1            |  | 4   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| 76        | INDEX FAST FULL SCAN                | SYS_C0020664                   |  | 1   | 37   |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| 77        | TABLE ACCESS FULL                   | PSP_SOURCE_SYSTEM_LAW_ASSOC    |  | 1   | 21   |
| 3 (0)     | 00:00:01                            |                                |  |     |      |
| * 78      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_PAYROLL_ITEM_INFO     |  | 1   | 16   |
| 3 (0)     | 00:00:01                            |                                |  |     |      |
| * 79      | INDEX RANGE SCAN                    | PSP_QBDT_PAYROLL_ITEM_INFO_FK1 |  | 1   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 80      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY_PMTTEMPLATE |  | 1   | 127  |
| 4 (0)     | 00:00:01                            |                                |  |     |      |
| * 81      | INDEX RANGE SCAN                    | PSP_COMPANY_AGENCY_PAYMENT_FK1 |  | 1   |      |
| 2 (0)     | 00:00:01                            |                                |  |     |      |
| * 82      | HASH JOIN                           |                                |  | 1   | 520  |
| 68911 (1) | 00:00:03                            |                                |  |     |      |
| 83        | NESTED LOOPS                        |                                |  | 1   | 499  |
| 68908 (1) | 00:00:03                            |                                |  |     |      |
| 84        | NESTED LOOPS                        |                                |  | 1   | 462  |
| 68906 (1) | 00:00:03                            |                                |  |     |      |
| 85        | NESTED LOOPS                        |                                |  | 1   | 370  |
| 68902 (1) | 00:00:03                            |                                |  |     |      |
| 86        | NESTED LOOPS                        |                                |  | 1   | 254  |
| 68898 (1) | 00:00:03                            |                                |  |     |      |
| 87        | NESTED LOOPS                        |                                |  | 1   | 199  |



|       |          |          |  |                                     |                                |  |   |     |
|-------|----------|----------|--|-------------------------------------|--------------------------------|--|---|-----|
| 68894 | (1)      | 00:00:03 |  |                                     |                                |  | 1 | 125 |
| 88    |          |          |  | NESTED LOOPS                        |                                |  |   |     |
| 68891 | (1)      | 00:00:03 |  |                                     |                                |  |   |     |
| * 89  |          |          |  | TABLE ACCESS FULL                   | PSP_COMPANY_LAW                |  | 1 | 109 |
| 68888 | (1)      | 00:00:03 |  |                                     |                                |  |   |     |
| * 90  |          |          |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_PAYROLL_ITEM_INFO     |  | 1 | 16  |
| 3 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 91  |          |          |  | INDEX RANGE SCAN                    | PSP_QBDT_PAYROLL_ITEM_INFO_FK1 |  | 1 |     |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| 92    |          |          |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY             |  | 1 | 74  |
| 3 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 93  |          |          |  | INDEX RANGE SCAN                    | SYS_C0020551                   |  | 1 |     |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 94  |          |          |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                    |  | 1 | 55  |
| 4 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 95  |          |          |  | INDEX RANGE SCAN                    | SYS_C0020424                   |  | 1 |     |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| 96    |          |          |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_AGENCY_PMTTEMPLATE |  | 1 | 116 |
| 4 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 97  |          |          |  | INDEX RANGE SCAN                    | PSP_COMPANY_AGENCY_PAYMENT_FK1 |  | 1 |     |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 98  |          |          |  | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE            |  | 1 | 92  |
| 4 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 99  |          |          |  | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK1        |  | 1 |     |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| * 100 |          |          |  | INDEX RANGE SCAN                    | SYS_C0020664                   |  | 1 | 37  |
| 2 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |
| 101   |          |          |  | TABLE ACCESS FULL                   | PSP_SOURCE_SYSTEM_LAW_ASSOC    |  | 1 | 21  |
| 3 (0) | 00:00:01 |          |  |                                     |                                |  |   |     |

Predicate Information (identified by operation id):

```

8 - access("COMPANYLAW4"."LAW_FK"="LAW7"."LAW_ID" AND "COMPANYAGE5"."PAYMENT_TEMPLATE_FK"="LAW7"."
PAYMENT_TEMPLATE_F
K")
12 - filter(("QBDTPAYROL6"."COMPANY_LAW_FK" IS NOT NULL AND "QBDTPAYROL6"."MODIFIED_DATE">=:4 AND
"QBDTPAYROL6"."IS_DELETED"=0))
13 - filter(("COMPANYLAW4"."ADDITIONAL_COMPANY_LAW_FK" IS NULL AND LNNVL("COMPANYLAW4"."MODIFIED_DATE">=:
1)))
14 - access("COMPANYLAW4"."COMPANY_LAW_SEQ"="QBDTPAYROL6"."COMPANY_LAW_FK")
16 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYLAW4"."COMPANY_AGENCY_FK")
17 - filter(LNNVL("COMPANYAGE5"."MODIFIED_DATE">=:2))
18 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYAGE5"."COMPANY_AGENCY_FK")
20 - access(ROWID=ROWID)
24 - access("SOURCESYST1"."LAW_FK"="COMPANYLAW4"."LAW_FK")
25 - filter(("SOURCESYST1"."SOURCE_SYSTEM_FK"="COMPANY2"."SOURCE_SYSTEM_CD" AND
NVL("COMPANY2"."IS_DG_DISASSOCIATED",0)=0))
26 - access("COMPANY2"."COMPANY_SEQ"="COMPANYAGE3"."COMPANY_FK")
27 - filter(("TAXCOMPANY0_1"."SERVICE_START_DATE" IS NOT NULL AND INTERNAL_FUNCTION("TAXCOMPANY0_1"."
STATUS_CD") AND
LNNVL("TAXCOMPANY0_1"."MODIFIED_DATE">=:3)))
28 - access("TAXCOMPANY0_1"."COMPANY_FK"="COMPANY2"."COMPANY_SEQ")
29 - access("TAXCOMPANY0"."TAX_COMPANY_SERVICE_INFO_SEQ"="TAXCOMPANY0_1"."COMPANY_SERVICE_SEQ")
34 - access("SOURCESYST1"."LAW_FK"="COMPANYLAW4"."LAW_FK")
35 - access("COMPANYLAW4"."LAW_FK"="LAW7"."LAW_ID" AND "COMPANYAGE5"."PAYMENT_TEMPLATE_FK"="LAW7"."
PAYMENT_TEMPLATE_F
K")
38 - filter("COMPANYAGE5"."MODIFIED_DATE">=:2)
40 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYAGE5"."COMPANY_AGENCY_FK")
41 - filter(("COMPANYLAW4"."ADDITIONAL_COMPANY_LAW_FK" IS NULL AND LNNVL("COMPANYLAW4"."MODIFIED_DATE">=:
1)))
42 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYLAW4"."COMPANY_AGENCY_FK")
44 - access(ROWID=ROWID)
48 - filter(("SOURCESYST1"."SOURCE_SYSTEM_FK"="COMPANY2"."SOURCE_SYSTEM_CD" AND
NVL("COMPANY2"."IS_DG_DISASSOCIATED",0)=0))
49 - access("COMPANY2"."COMPANY_SEQ"="COMPANYAGE3"."COMPANY_FK")
50 - filter(("TAXCOMPANY0_1"."SERVICE_START_DATE" IS NOT NULL AND INTERNAL_FUNCTION("TAXCOMPANY0_1"."

```

```

STATUS_CD")))
51 - access("TAXCOMPANY0_1"."COMPANY_FK"="COMPANY2"."COMPANY_SEQ")
52 - access("TAXCOMPANY0"."TAX_COMPANY_SERVICE_INFO_SEQ"="TAXCOMPANY0_1"."COMPANY_SERVICE_SEQ")
53 - filter("QBDTPAYROL6"."IS_DELETED"=0)
54 - access("COMPANYLAW4"."COMPANY_LAW_SEQ"="QBDTPAYROL6"."COMPANY_LAW_FK")
 filter("QBDTPAYROL6"."COMPANY_LAW_FK" IS NOT NULL)
55 - access("ITEM_2"="LAW7"."LAW_ID" AND "ITEM_1"="LAW7"."PAYMENT_TEMPLATE_FK")
57 - access(ROWID=ROWID)
64 - access("SOURCESYST1"."SOURCE_SYSTEM_FK"="COMPANY2"."SOURCE_SYSTEM_CD" AND
 "SOURCESYST1"."LAW_FK"="COMPANYLAW4"."LAW_FK")
65 - access("TAXCOMPANY0"."TAX_COMPANY_SERVICE_INFO_SEQ"="TAXCOMPANY0_1"."COMPANY_SERVICE_SEQ")
69 - filter(("TAXCOMPANY0_1"."SERVICE_START_DATE" IS NOT NULL AND "TAXCOMPANY0_1"."MODIFIED_DATE">=:3 AND
 INTERNAL_FUNCTION("TAXCOMPANY0_1"."STATUS_CD")))
70 - filter(NVL("COMPANY2"."IS_DG_DISASSOCIATED",0)=0)
71 - access("TAXCOMPANY0_1"."COMPANY_FK"="COMPANY2"."COMPANY_SEQ")
73 - access("COMPANY2"."COMPANY_SEQ"="COMPANYAGE3"."COMPANY_FK")
74 - filter(("COMPANYLAW4"."ADDITIONAL_COMPANY_LAW_FK" IS NULL AND LNNVL("COMPANYLAW4"."MODIFIED_DATE">=:
1)))
75 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYLAW4"."COMPANY_AGENCY_FK")
78 - filter("QBDTPAYROL6"."IS_DELETED"=0)
79 - access("COMPANYLAW4"."COMPANY_LAW_SEQ"="QBDTPAYROL6"."COMPANY_LAW_FK")
 filter("QBDTPAYROL6"."COMPANY_LAW_FK" IS NOT NULL)
80 - filter(LNNVL("COMPANYAGE5"."MODIFIED_DATE">=:2))
81 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYAGE5"."COMPANY_AGENCY_FK")
82 - access("SOURCESYST1"."SOURCE_SYSTEM_FK"="COMPANY2"."SOURCE_SYSTEM_CD" AND
 "SOURCESYST1"."LAW_FK"="COMPANYLAW4"."LAW_FK")
89 - filter(("COMPANYLAW4"."MODIFIED_DATE">=:1 AND "COMPANYLAW4"."ADDITIONAL_COMPANY_LAW_FK" IS NULL))
90 - filter("QBDTPAYROL6"."IS_DELETED"=0)
91 - access("COMPANYLAW4"."COMPANY_LAW_SEQ"="QBDTPAYROL6"."COMPANY_LAW_FK")
 filter("QBDTPAYROL6"."COMPANY_LAW_FK" IS NOT NULL)
93 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYLAW4"."COMPANY_AGENCY_FK")
94 - filter(NVL("COMPANY2"."IS_DG_DISASSOCIATED",0)=0)
95 - access("COMPANY2"."COMPANY_SEQ"="COMPANYAGE3"."COMPANY_FK")
97 - access("COMPANYAGE3"."COMPANY_AGENCY_SEQ"="COMPANYAGE5"."COMPANY_AGENCY_FK")
98 - filter(("TAXCOMPANY0_1"."SERVICE_START_DATE" IS NOT NULL AND INTERNAL_FUNCTION("TAXCOMPANY0_1"."
STATUS_CD")))
99 - access("TAXCOMPANY0_1"."COMPANY_FK"="COMPANY2"."COMPANY_SEQ")
100 - access("TAXCOMPANY0"."TAX_COMPANY_SERVICE_INFO_SEQ"="TAXCOMPANY0_1"."COMPANY_SERVICE_SEQ")

```

Note

-----

- this is an adaptive plan

## Opportunity #11

### DailyBatchJobsProcessor - 2nd0g8cwbz6zg

```
BEGIN prc_offload_update_payroll(:1 , :2 , :3 , :4); END;
```

Note : Opportunity #12 is part of this procedure.

## Opportunity #12

### DailyBatchJobsProcessor - 1w9qy5w5y3cxf

```

UPDATE
 PSP_PAYROLL_RUN PR
SET PAYROLL_RUN_STATUS = 'OffloadedAll',
 VERSION = VERSION + 1,
 MODIFIER_ID = : B2,
 MODIFIED_DATE = : B1
WHERE EXISTS(
 SELECT 'T'

```

```

FROM PSP_MONEY_MOVEMENT_TRANSACTION MMT1,
 PSP_FINANCIAL_TRANSACTION FT1
WHERE FT1.PAYROLL_RUN_FK = PR.PAYROLL_RUN_SEQ
 AND MMT1.OFFLOAD_BATCH_FK = : B4
 AND MMT1.INITIATION_DATE = : B3
 AND MMT1.MONEY_MOVEMENT_TRANSACTION_SEQ = FT1.MONEY_MOVEMENT_TRANSACTION_FK
 AND FT1.SETTLEMENT_DATE >= MMT1.INITIATION_DATE
 AND TRUNC(FT1.SETTLEMENT_DATE) >= TRUNC(MMT1.INITIATION_DATE)
 AND FT1.CURRENT_TRANSACTION_STATE_FK = 'Executed'
 AND FT1.TRANSACTION_TYPE_FK = 'EmployeeDdCredit'
)

```

Plan hash value: 730446350

| Id | Operation                                 | Name                           | Rows | Bytes |
|----|-------------------------------------------|--------------------------------|------|-------|
| 0  | UPDATE STATEMENT                          |                                |      |       |
| 1  | UPDATE                                    | PSP_PAYROLL_RUN                |      |       |
| 2  | NESTED LOOPS                              |                                | 1    | 222   |
| 3  | VIEW                                      | VW_SQ_1                        | 1    | 129   |
| 4  | SORT UNIQUE                               |                                | 1    | 197   |
| 5  | NESTED LOOPS                              |                                | 1    | 197   |
| 6  | NESTED LOOPS                              |                                | 13   | 197   |
| 7  | PARTITION RANGE SINGLE                    |                                | 13   | 1105  |
| 8  | TABLE ACCESS BY LOCAL INDEX ROWID BATCHED | PSP_MONEY_MOVEMENT_TRANSACTION | 13   | 1105  |
| 9  | INDEX RANGE SCAN                          | PSP_MONEY_MOVEMENT_TRANSAC_FK3 | 5216 |       |
| 10 | PARTITION RANGE AND                       |                                | 1    |       |
| 11 | INDEX RANGE SCAN                          | PSP_FINANCIAL_TRANSACTION_FK10 | 1    |       |
| 12 | TABLE ACCESS BY LOCAL INDEX ROWID         | PSP_FINANCIAL_TRANSACTION      | 1    | 112   |
| 13 | INDEX RANGE SCAN                          | SYS_C0020494                   | 1    | 93    |

Predicate Information (identified by operation id):

```

8 - filter("MMT1"."INITIATION_DATE"=:B3)
9 - access("MMT1"."OFFLOAD_BATCH_FK"=:B4)
11 - access("MMT1"."MONEY_MOVEMENT_TRANSACTION_SEQ"="FT1"."MONEY_MOVEMENT_TRANSACTION_FK")
12 - filter(("FT1"."SETTLEMENT_DATE">=:B3 AND "FT1"."CURRENT_TRANSACTION_STATE_FK"='Executed' AND
"FT1"."TRANSACTION_TYPE_FK"='EmployeeDdCredit' AND TRUNC(INTERNAL_FUNCTION("FT1"."
SETTLEMENT_DATE"))>=TRUNC(INTERNAL_FUNCTION("MMT1"."INITI
ATION_DATE"))) AND "FT1"."SETTLEMENT_DATE">="MMT1"."INITIATION_DATE"))
13 - access("ITEM_1"="PR"."PAYROLL_RUN_SEQ")

```

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                        | SQL Text                          |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|-----------------------------------|-----------------------------------|
| 85,082,230     | 1          | 85,082,230.00  | 84.92   | 1,505.50         | 62.41 | 38.50 | 6qk524uazdf0c | GemsAccountsReceivableProcess or: | UPDATE PSP_FINANCIAL_TRANS_STA... |
| 2,793,936      | 1          | 2,793,936.00   | 2.79    | 131.11           | 41.74 | 67.78 | 5vfmc964z0tqr | ViewMyPaycheckAdapter:232         | /* criteria query */ select co... |
| 1,289,960      | 1          | 1,289,960.00   | 1.29    | 332.87           | 27.91 | 76.05 | 2su4my5wrmmbv | ViewMyPaycheckAdapter:199         | select * from ( /* criteria qu... |

## Opportunity #13

### ViewMyPaycheckAdapter - 5vfmc964z0tqr

```

/* criteria query */
select count(this_.EMPLOYEE_SEQ) as y0_
from PSP_EMPLOYEE this_
 inner join PSP_INDIVIDUAL this_1_ on this_.EMPLOYEE_SEQ = this_1_.INDIVIDUAL_SEQ
where (
 exists(
 /* criteria query */
 select d1_.COMPANY_SERVICE_SEQ as y0_
 from PSP_COMPANY_SERVICE d1_
 left outer join PSP_DDCOMPANY_SERVICE_INFO d1_1_
 on d1_.COMPANY_SERVICE_SEQ = d1_1_.DDCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_CDCOMPANY_SERVICE_INFO d1_2_
 on d1_.COMPANY_SERVICE_SEQ = d1_2_.CDCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_BPCOMPANY_SERVICE_INFO d1_3_
 on d1_.COMPANY_SERVICE_SEQ = d1_3_.BPCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_TP401KCOMPANY_SERVICE_INFO d1_4_
 on d1_.COMPANY_SERVICE_SEQ = d1_4_.TP401KCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_RACOMPANY_SERVICE_INFO d1_5_
 on d1_.COMPANY_SERVICE_SEQ = d1_5_.RACOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_TAX_COMPANY_SERVICE_INFO d1_6_ on
d1_.COMPANY_SERVICE_SEQ = d1_6_.TAX_COMPANY_SERVICE_INFO_SEQ
 inner join PSP_SERVICE t0x1_ on d1_.SERVICE_FK = t0x1_.SERVICE_CD
 where
 t0x1_.SERVICE_CD = : 1
 and d1_.COMPANY_FK = this_.COMPANY_FK
)
 and lower(this_1_.EMAIL) like : 2

```

Plan hash value: 3174870401

| Id                        | Operation             | Name           | Rows  | Bytes |
|---------------------------|-----------------------|----------------|-------|-------|
| TempSpc                   | Cost (%CPU)           | Time           |       |       |
| 0                         | SELECT STATEMENT      |                |       |       |
| 985K(100)                 |                       |                |       |       |
| 1                         | SORT AGGREGATE        |                | 1     | 631   |
| 2                         | HASH JOIN             |                | 96090 | 57M   |
| 194M  985K (1)  00:00:39  |                       |                |       |       |
| 3                         | TABLE ACCESS FULL     | PSP_INDIVIDUAL | 3586K | 153M  |
| 445K (1)  00:00:18        |                       |                | 1557K | 870M  |
| 4                         | NESTED LOOPS          |                |       |       |
| 486K (1)  00:00:19        |                       |                | 1582K | 870M  |
| 5                         | NESTED LOOPS          |                |       |       |
| 486K (1)  00:00:19        |                       |                |       |       |
| 6                         | VIEW                  | VW_SQ_1        | 56518 | 27M   |
| 50771 (1)  00:00:02       |                       |                | 56518 | 42M   |
| 7                         | HASH UNIQUE           |                |       |       |
| 176M  50771 (1)  00:00:02 |                       |                |       |       |
| 8                         | HASH JOIN RIGHT OUTER |                | 226K  | 170M  |
| 31419 (1)  00:00:02       |                       |                |       |       |

|       |       |                                     |                            |       |       |
|-------|-------|-------------------------------------|----------------------------|-------|-------|
| 9     |       | INDEX FAST FULL SCAN                | SYS_C0020567               | 733   | 27121 |
| 3     | (0)   | 00:00:01                            |                            |       |       |
| 10    |       | HASH JOIN RIGHT OUTER               |                            | 226K  | 162M  |
| 31415 | (1)   | 00:00:02                            |                            |       |       |
| 11    |       | INDEX FAST FULL SCAN                | SYS_C0020582               | 141   | 5217  |
| 2     | (0)   | 00:00:01                            |                            |       |       |
| 12    |       | HASH JOIN RIGHT OUTER               |                            | 226K  | 154M  |
| 31413 | (1)   | 00:00:02                            |                            |       |       |
| 13    |       | INDEX FULL SCAN                     | SYS_C0020630               | 1     | 512   |
| 0     | (0)   |                                     |                            |       |       |
| 14    |       | HASH JOIN OUTER                     |                            | 226K  | 44M   |
| 38M   | 31412 | (1)                                 | 00:00:02                   |       |       |
| 15    |       | HASH JOIN RIGHT OUTER               |                            | 223K  | 36M   |
| 6520K | 23876 | (1)                                 | 00:00:01                   |       |       |
| 16    |       | INDEX FAST FULL SCAN                | SYS_C0020664               | 136K  | 4918K |
| 418   | (1)   | 00:00:01                            |                            |       |       |
| 17    |       | HASH JOIN RIGHT OUTER               |                            | 223K  | 28M   |
| 9208K | 21619 | (1)                                 | 00:00:01                   |       |       |
| 18    |       | TABLE ACCESS FULL                   | PSP_BPCOMPANY_SERVICE_INFO | 192K  | 6950K |
| 451   | (1)   | 00:00:01                            |                            |       |       |
| 19    |       | MERGE JOIN CARTESIAN                |                            | 223K  | 20M   |
| 19591 | (1)   | 00:00:01                            |                            |       |       |
| 20    |       | INDEX RANGE SCAN                    | SYS_C0020516               | 1     | 12    |
| 1     | (0)   | 00:00:01                            |                            |       |       |
| 21    |       | BUFFER SORT                         |                            | 223K  | 17M   |
| 19590 | (1)   | 00:00:01                            |                            |       |       |
| 22    |       | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE        | 223K  | 17M   |
| 19590 | (1)   | 00:00:01                            |                            |       |       |
| 23    |       | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK2    | 223K  |       |
| 754   | (1)   | 00:00:01                            |                            |       |       |
| 24    |       | TABLE ACCESS FULL                   | PSP_DDCOMPANY_SERVICE_INFO | 1223K | 43M   |
| 2781  | (1)   | 00:00:01                            |                            |       |       |
| 25    |       | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2           | 28    |       |
| 3     | (0)   | 00:00:01                            |                            |       |       |
| 26    |       | TABLE ACCESS BY INDEX ROWID         | PSP_EMPLOYEE               | 28    | 2072  |
| 18    | (0)   | 00:00:01                            |                            |       |       |

Note

- this is an adaptive plan

## Opportunity #14

### ViewMyPaycheckAdapter - 2su4my5wrmbv

```

select *
from (

 /* criteria query */
 select this_.EMPLOYEE_SEQ as INDIVIDUAL1_207_1_,
 this_1_.VERSION as VERSION207_1_,
 this_1_.CREATOR_ID as CREATOR3_207_1_,
 this_1_.CREATED_DATE as CREATED4_207_1_,
 this_1_.MODIFIER_ID as MODIFIER5_207_1_,
 this_1_.MODIFIED_DATE as MODIFIED6_207_1_,
 this_1_.REALM_ID as REALM7_207_1_,
 this_1_.HAS_INVALID_EMAIL as HAS8_207_1_,
 this_1_.FIRST_NAME as FIRST9_207_1_,
 this_1_.GENDER_CD as GENDER10_207_1_,
 this_1_.LAST_NAME as LAST11_207_1_,
 this_1_.MIDDLE_NAME as MIDDLE12_207_1_,
 this_1_.COMMUNICATION_TYPE_PREFERENCE as COMMUNI13_207_1_,
 this_1_.EMAIL as EMAIL207_1_,
 this_1_.PHONE as PHONE207_1_,

```

|                                        |                       |
|----------------------------------------|-----------------------|
| this_1_.SUFFIX                         | as SUFFIX207_1_       |
| this_1_.MAILING_ADDRESS_FK             | as MAILING17_207_1_   |
| this_.IS_VIEWING_PAYSTUB_DISABLED      | as IS2_285_1_         |
| this_.SOURCE_EMPLOYEE_ID               | as SOURCE3_285_1_     |
| this_.STATUS_CD                        | as STATUS4_285_1_     |
| this_.STATUS_EFFECTIVE_DATE            | as STATUS5_285_1_     |
| this_.HIRE_DATE                        | as HIRE6_285_1_       |
| this_.RE_HIRE_DATE                     | as RE7_285_1_         |
| this_.FED_FILING_STATUS                | as FED8_285_1_        |
| this_.WORK_STATE                       | as WORK9_285_1_       |
| this_.TERMINATION_DATE                 | as TERMINA10_285_1_   |
| this_.FED_ALLOWANCES                   | as FED11_285_1_       |
| this_.IS_STATUTORY                     | as IS12_285_1_        |
| this_.HAS_RETIREMENT_PLAN              | as HAS13_285_1_       |
| this_.HAS_THIRD_PARTY_SICK_PAY         | as HAS14_285_1_       |
| this_.IS_DECEASED                      | as IS15_285_1_        |
| this_.QUALIFIES_FOR_AEIC               | as QUALIFIES16_285_1_ |
| this_.FED_EXTRA_WITHHOLDING            | as FED17_285_1_       |
| this_.LIVE_STATE                       | as LIVE18_285_1_      |
| this_.PAY_PERIOD                       | as PAY19_285_1_       |
| this_.FED_CLAIM_DEPENDENTS             | as FED20_285_1_       |
| this_.FED_OTHER_INCOME                 | as FED21_285_1_       |
| this_.FED_DEDUCTIONS                   | as FED22_285_1_       |
| this_.FED_MULTIPLE_JOBS                | as FED23_285_1_       |
| this_.FED_W4_EMPLOYEE_PREF             | as FED24_285_1_       |
| this_.CONSUMER_REALM_ID                | as CONSUMER25_285_1_  |
| this_.BIRTH_DATE_ENC                   | as BIRTH26_285_1_     |
| this_.TAX_ID_ENC                       | as TAX27_285_1_       |
| this_.PERSONA_ID                       | as PERSONA28_285_1_   |
| this_.IS_DG_DISASSOCIATED              | as IS29_285_1_        |
| this_.PUBLISH_STATUS                   | as PUBLISH30_285_1_   |
| this_.TP_401K_INFO_IS_HCE              | as TP31_285_1_        |
| this_.TP_401K_INFO_OWNER_PERCENT       | as TP32_285_1_        |
| this_.TP_401K_INFO_IS_FAMILY_MEMBER    | as TP33_285_1_        |
| this_.TP_401K_INFO_LAST_UPLOAD_DATE    | as TP34_285_1_        |
| this_.COMPANY_FK                       | as COMPANY35_285_1_   |
| company2_.COMPANY_SEQ                  | as COMPANY1_194_0_    |
| company2_.VERSION                      | as VERSION194_0_      |
| company2_.CREATOR_ID                   | as CREATOR3_194_0_    |
| company2_.CREATED_DATE                 | as CREATED4_194_0_    |
| company2_.MODIFIER_ID                  | as MODIFIER5_194_0_   |
| company2_.MODIFIED_DATE                | as MODIFIED6_194_0_   |
| company2_.REALM_ID                     | as REALM7_194_0_      |
| company2_.PRIVATE_KEY_ENC              | as PRIVATE8_194_0_    |
| company2_.PHONE                        | as PHONE194_0_        |
| company2_.DBA_NAME                     | as DBA10_194_0_       |
| company2_.LEGAL_NAME                   | as LEGAL11_194_0_     |
| company2_.SOURCE_COMPANY_ID            | as SOURCE12_194_0_    |
| company2_.DEBUG_LOGGING                | as DEBUG13_194_0_     |
| company2_.NOTIFICATION_EMAIL           | as NOTIFIC14_194_0_   |
| company2_.NEXT_PAYROLL_TRANSACTION_ID  | as NEXT15_194_0_      |
| company2_.NEXT_PAYCHECK_ID             | as NEXT16_194_0_      |
| company2_.NEXT_EMPLOYEE_ID             | as NEXT17_194_0_      |
| company2_.NEXT_PAYROLL_ITEM_ID         | as NEXT18_194_0_      |
| company2_.ACCOUNT_LOCKED_UNTIL         | as ACCOUNT19_194_0_   |
| company2_.NBR_OF_FAILED_LOGIN_ATTEMPTS | as NBR20_194_0_       |
| company2_.CURRENT_TOKEN                | as CURRENT21_194_0_   |
| company2_.SOURCE_SYSTEM_CD             | as SOURCE22_194_0_    |
| company2_.TAX_EXEMPT_EXPIRATION_DATE   | as TAX23_194_0_       |
| company2_.NAME_CONTROL                 | as NAME24_194_0_      |
| company2_.IS_FLAGGED_FOR_FRAUD         | as IS25_194_0_        |
| company2_.SIGN_UP_DATE                 | as SIGN26_194_0_      |
| company2_.PUBLIC_KEY                   | as PUBLIC27_194_0_    |
| company2_.NBR_FAILED_AUTHENTICATIONS   | as NBR28_194_0_       |
| company2_.CLOUD_CURRENT_TOKEN          | as CLOUD29_194_0_     |
| company2_.I_A_M_REALM_ID               | as I30_194_0_         |
| company2_.TAX_EXEMPT_STATUS            | as TAX31_194_0_       |
| company2_.PRICE_TYPE                   | as PRICE32_194_0_     |
| company2_.FED_TAX_ID_ENC               | as FED33_194_0_       |
| company2_.D_D_PUBLISH_FLAG             | as D34_194_0_         |
| company2_.O_I_I_FLAG                   | as O35_194_0_         |

```

company2_.IS_DG_DISASSOCIATED as IS36_194_0_,
company2_.PUBLISH_STATUS as PUBLISH37_194_0_,
company2_.COMPLIANCE_AD DRESS_FK as COMPLIANCE38_194_0_,
company2_.MAILING_ADDRESS_FK as MAILING39_194_0_,
company2_.LEGAL_ADDRESS_FK as LEGAL40_194_0_,
company2_.PAYROLL_FREQUENCY_FK as PAYROLL41_194_0_,
company2_.OFFLOAD_GROUP_FK as OFFLOAD42_194_0_,
company2_.FUNDING_MODEL_FK as FUNDING43_194_0_
from PSP_EMPLOYEE this_
 inner join PSP_INDIVIDUAL this_1_ on this_.EMPLOYEE_SEQ = this_1_.INDIVIDUAL_SEQ
 left outer join PSP_COMPANY company2_ on this_.COMPANY_FK = company2_.COMPANY_SEQ
where (
 exists(
 /* criteria query */
 select d1_.COMPANY_SERVICE_SEQ as y0_
 from PSP_COMPANY_SERVICE d1_
 left outer join PSP_DDCOMPANY_SERVICE_INFO d1_1_
 on d1_.COMPANY_SERVICE_SEQ = d1_1_.
DDCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_BPCOMPANY_SERVICE_INFO d1_2_
 on d1_.COMPANY_SERVICE_SEQ = d1_2_.
BPCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_CDCOMPANY_SERVICE_INFO d1_3_
 on d1_.COMPANY_SERVICE_SEQ = d1_3_.
CDCOMPANY_SERVICE_INFO_SEQ
 left outer join PSP_TP401KCOMPANY_SERVICE_INFO d1_4_
 on d1_.COMPANY_SERVICE_SEQ = d1_4_.TP401KCOMPANY_SERVIC
E_INFO_SEQ
 left outer join PSP_TAX_COMPANY_SERVICE_INFO d1_5_ on
d1_.COMPANY_SERVICE_SEQ = d1_5_.TAX_COMPANY_SERVICE_INFO_SEQ
 left outer join PSP_RACOMPANY_SERVICE_INFO d1_6_ on
d1_.COMPANY_SERVICE_SEQ = d1_6_.RACOMPANY_SERVICE_INFO_SEQ
 inner join PSP_SERVICE t0x1_ on d1_.SERVICE_FK = t0x1_.SERVICE_CD
 where
 t0x1_.SERVICE_CD = : 1
 and d1_.COMPANY_FK = this_.COMPANY_FK
)
 and lower(this_1_.EMAIL) like : 2
)
 order by this_1_.EMAIL asc
)
where rownum <= : 3

```

Plan hash value: 2810162369

| Id | Operation             | Name    | Rows  | Bytes |
|----|-----------------------|---------|-------|-------|
| 0  | SELECT STATEMENT      |         |       |       |
| 1  | COUNT STOPKEY         |         |       |       |
| 2  | VIEW                  |         | 97908 |       |
| 3  | SORT ORDER BY STOPKEY |         | 97908 | 112M  |
| 4  | NESTED LOOPS OUTER    |         | 97908 |       |
| 5  | HASH JOIN             |         | 96090 | 42M   |
| 6  | NESTED LOOPS          |         | 1557K |       |
| 7  | NESTED LOOPS          |         | 1582K |       |
| 8  | VIEW                  | Vw_SQ_1 | 56518 |       |

|              |                                     |                            |       |       |
|--------------|-------------------------------------|----------------------------|-------|-------|
| 9            | HASH UNIQUE                         |                            | 56518 | 42M   |
| 176M  50771  | (1)  00:00:02                       |                            |       |       |
| 10           | HASH JOIN RIGHT OUTER               |                            | 226K  |       |
| 170M         | 31419 (1)  00:00:02                 |                            |       |       |
| 11           | INDEX FAST FULL SCAN                | SYS_C0020567               | 733   | 27121 |
|              | 3 (0)  00:00:01                     |                            |       |       |
| 12           | HASH JOIN RIGHT OUTER               |                            | 226K  |       |
| 162M         | 31415 (1)  00:00:02                 |                            |       |       |
| 13           | INDEX FAST FULL SCAN                | SYS_C0020582               | 141   | 5217  |
|              | 2 (0)  00:00:01                     |                            |       |       |
| 14           | HASH JOIN RIGHT OUTER               |                            | 226K  |       |
| 154M         | 31413 (1)  00:00:02                 |                            |       |       |
| 15           | INDEX FULL SCAN                     | SYS_C0020630               | 1     | 512   |
|              | 0 (0)                               |                            |       |       |
| 16           | HASH JOIN OUTER                     |                            | 226K  | 44M   |
| 38M  31412   | (1)  00:00:02                       |                            |       |       |
| 17           | HASH JOIN RIGHT OUTER               |                            | 223K  | 36M   |
| 6520K  23876 | (1)  00:00:01                       |                            |       |       |
| 18           | INDEX FAST FULL SCAN                | SYS_C0020664               | 136K  |       |
| 4918K        | 418 (1)  00:00:01                   |                            |       |       |
| 19           | HASH JOIN RIGHT OUTER               |                            | 223K  | 28M   |
| 9208K  21619 | (1)  00:00:01                       |                            |       |       |
| 20           | TABLE ACCESS FULL                   | PSP_BPCOMPANY_SERVICE_INFO | 192K  |       |
| 6950K        | 451 (1)  00:00:01                   |                            |       |       |
| 21           | MERGE JOIN CARTESIAN                |                            | 223K  |       |
| 20M          | 19591 (1)  00:00:01                 |                            |       |       |
| 22           | INDEX RANGE SCAN                    | SYS_C0020516               | 1     | 12    |
|              | 1 (0)  00:00:01                     |                            |       |       |
| 23           | BUFFER SORT                         |                            | 223K  |       |
| 17M          | 19590 (1)  00:00:01                 |                            |       |       |
| 24           | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_SERVICE        | 223K  |       |
| 17M          | 19590 (1)  00:00:01                 |                            |       |       |
| 25           | INDEX RANGE SCAN                    | PSP_COMPANY_SERVICE_FK2    | 223K  |       |
|              | 754 (1)  00:00:01                   |                            |       |       |
| 26           | TABLE ACCESS FULL                   | PSP_DDCOMPANY_SERVICE_INFO | 1223K |       |
| 43M          | 2781 (1)  00:00:01                  |                            |       |       |
| 27           | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2           | 28    |       |
|              | 3 (0)  00:00:01                     |                            |       |       |
| 28           | TABLE ACCESS BY INDEX ROWID         | PSP_EMPLOYEE               | 28    | 7392  |
|              | 18 (0)  00:00:01                    |                            |       |       |
| 29           | TABLE ACCESS FULL                   | PSP_INDIVIDUAL             | 3586K |       |
| 550M         | 445K (1)  00:00:18                  |                            |       |       |
| 30           | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                | 1     | 739   |
|              | 4 (0)  00:00:01                     |                            |       |       |
| 31           | INDEX RANGE SCAN                    | SYS_C0020424               | 1     |       |
|              | 2 (0)  00:00:01                     |                            |       |       |

Note

- this is an adaptive plan

SQL ordered by Reads (23-Sep-21 18:30 - 23-Sep-21 19:00 )

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                     | SQL Text                          |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|--------------------------------|-----------------------------------|
| 4,715,710      | 1          | 4,715,710.00   | 12.35   | 99.83            | 19.88 | 42.17 | 39f2687k24czy | BaseATFExtractFileProcess: 146 | /* Select companyDailyLiabilit... |
| 1,503,359      | 1          | 1,503,359.00   | 3.94    | 860.18           | 15.24 | 91.30 | 2nd0g8cwbz6zg | DailyBatchJobsProcessor: 776   | BEGIN prc_offload_update_payro... |
| 1,429,802      | 1          | 1,429,802.00   | 3.74    | 792.92           | 12.71 | 94.02 | 1w9qy5w5y3cxf | DailyBatchJobsProcessor: 776   | UPDATE PSP_PAYROLL_RUN PR SET ... |
| 591,873        | 1          | 591,873.00     | 1.55    | 338.75           | 14.63 | 89.22 | 9zdchramsws6b | DailyBatchJobsProcessor: 776   | INSERT /*+ APPEND */ INTO PSP_... |
| 591,873        | 1          | 591,873.00     | 1.55    | 338.75           | 14.63 | 89.22 | fx6bn8avj7cp2 | DailyBatchJobsProcessor: 776   | BEGIN PRC_OFFLOAD_INSERT_FTS(...  |
| 498,713        | 1          | 498,713.00     | 1.31    | 86.50            | 5.10  | 95.87 | 25kf4y2hsgf1x | BaseATFExtractFileProcess: 146 | /* select c.SourceCompanyId, c... |
| 405,821        | 1          | 405,821.00     | 1.06    | 75.59            | 19.96 | 89.15 | bbd4wbqnzr8p6 | ATFDataExtractProcessor: 497   | /* named HQL query findCompani... |



## Opportunity #15

### BaseATFExtractFileProcess - 39f2687k24czy

```

/* Select companyDailyLiability, companyDailyLiability.Company.SourceCompanyId, companyDailyLiability.
Company.SourceSystemCd
from com.intuit.sbd.payroll.psp.domain.CompanyDailyLiability as companyDailyLiability
where companyDailyLiability.LiabilityDate <= :endDate and companyDailyLiability.ModifiedDate >= :lastRunDate
order by companyDailyLiability.Company, companyDailyLiability.LiabilityDate, companyDailyLiability.Law */
select companydai0_.COMPANY_DAILY_LIABILITY_SEQ as col_0_0_,
 company1_.SOURCE_COMPANY_ID as col_1_0_,
 company1_.SOURCE_SYSTEM_CD as col_2_0_,
 companydai0_.COMPANY_DAILY_LIABILITY_SEQ as COMPANY1_31_,
 companydai0_.VERSION as VERSION31_,
 companydai0_.CREATOR_ID as CREATOR3_31_,
 companydai0_.CREATED_DATE as CREATED4_31_,
 companydai0_.MODIFIER_ID as MODIFIER5_31_,
 companydai0_.MODIFIED_DATE as MODIFIED6_31_,
 companydai0_.REALM_ID as REALM7_31_,
 companydai0_.LIABILITY_DATE as LIABILITY8_31_,
 companydai0_.TAXABLE_WAGES as TAXABLE9_31_,
 companydai0_.TOTAL_WAGES as TOTAL10_31_,
 companydai0_.TOTAL_TIPS_AMOUNT as TOTAL11_31_,
 companydai0_.TAX_AMOUNT as TAX12_31_,
 companydai0_.LAW_FK as LAW13_31_,
 companydai0_.COMPANY_FK as COMPANY14_31_
from PSP_COMPANY_DAILY_LIABILITY companydai0_,
 PSP_COMPANY company1_
where companydai0_.COMPANY_FK = company1_.COMPANY_SEQ
and companydai0_.LIABILITY_DATE <= : 1
and companydai0_.MODIFIED_DATE >=: 2
order by
 companydai0_.COMPANY_FK,
 companydai0_.LIABILITY_DATE,
 companydai0_.LAW_FK

```

Plan hash value: 2440059379

| Id | Operation                   | Name                        | Rows | Bytes | Cost (%CPU) | Time     |
|----|-----------------------------|-----------------------------|------|-------|-------------|----------|
| 0  | SELECT STATEMENT            |                             |      |       | 1164K(100)  |          |
| 1  | SORT ORDER BY               |                             | 8    | 1816  | 1164K (1)   | 00:00:46 |
| 2  | NESTED LOOPS                |                             | 8    | 1816  | 1164K (1)   | 00:00:46 |
| 3  | NESTED LOOPS                |                             | 8    | 1816  | 1164K (1)   | 00:00:46 |
| 4  | TABLE ACCESS FULL           | PSP_COMPANY_DAILY_LIABILITY | 8    | 1400  | 1164K (1)   | 00:00:46 |
| 5  | INDEX RANGE SCAN            | SYS_C0020424                | 1    |       | 2 (0)       | 00:00:01 |
| 6  | TABLE ACCESS BY INDEX ROWID | PSP_COMPANY                 | 1    | 52    | 3 (0)       | 00:00:01 |

SQL ordered by Reads (22-Sep-21 19:00 - 22-Sep-21 19:45)

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                        | SQL Text                          |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|-----------------------------------|-----------------------------------|
| 44,813,170     | 0          |                | 52.76   | 319.75           | 46.19 | 54.83 | 6qk524uazdf0c | GemsAccountsReceivableProcess or: | UPDATE PSP_FINANCIAL_TRANS_STA... |
| 4,713,495      | 1          | 4,713,495.00   | 5.55    | 99.97            | 17.71 | 38.75 | fud06g9jzyyh2 | BaseATFExtractFileProcess:146     | /* Select companyDailyLiabilit... |
| 3,289,382      | 1          | 3,289,382.00   | 3.87    | 156.61           | 9.89  | 68.46 | 5nmd8a96rcf9j | BaseATFExtractFileProcess:146     | /* Select ptp.Law , ptp.Amount... |
| 2,324,485      | 1          | 2,324,485.00   | 2.74    | 282.61           | 7.75  | 94.36 | 5ttmr2d161kga | BaseATFExtractFileProcess:146     | /* select co, serviceInfo.Stat... |
| 1,903,416      | 1          | 1,903,416.00   | 2.24    | 1,236.51         | 16.76 | 86.97 | 9zdcchrmsws6b | DailyBatchJobsProcessor:776       | INSERT /*+ APPEND */ INTO PSP_... |

|           |   |              |      |          |       |       |               |                               |                                   |
|-----------|---|--------------|------|----------|-------|-------|---------------|-------------------------------|-----------------------------------|
| 1,903,416 | 1 | 1,903,416.00 | 2.24 | 1,236.51 | 16.76 | 86.97 | fx6bn8avj7cp2 | DailyBatchJobsProcessor:776   | BEGIN PRC_OFFLOAD_INSERT_FTS (... |
| 1,300,244 | 1 | 1,300,244.00 | 1.53 | 161.06   | 5.44  | 88.49 | a1zfnguds0wgx | BaseATFExtractFileProcess:146 | /* select comp.SourceCompanyId... |

Opportunity #16

DailyBatchJobsProcessor - 9zdchramsws6b

```
INSERT
/*+ APPEND */
INTO PSP_FINANCIAL_TRANS_STATE (FINANCIAL_TRANS_STATE_SEQ, VERSION,
 CREATOR_ID, CREATED_DATE, MODIFIER_ID,
 MODIFIED_DATE, REALM_ID, TRANSACTION_STATE_EFF_DATE,
 INSERT_USER_ID, GEMS_UPLOAD_BATCH_FK,
 FINANCIAL_TRANSACTION_FK, TRANSACTION_STATE_FK,
 TRANSACTION_RESPONSE_FK, COMPANY_FK,
 TRANSACTION_TYPE_FK) (
 SELECT FN_FORMAT_SYSGUID(
 SYS_GUID()
),
 0,
 : B5,
 : B4,
 : B5,
 : B4,
 -1,
 TRUNC(
 NEW_TIME(: B3, 'GMT', 'PDT')
) + .5,
 NULL,
 NULL,
 FT.FINANCIAL_TRANSACTION_SEQ,
 'Executed',
 NULL,
 FT.COMPANY_FK,
 FT.TRANSACTION_TYPE_FK
 FROM PSP_FINANCIAL_TRANSACTION FT,
 PSP_MONEY_MOVEMENT_TRANSACTION MMT
 WHERE MMT.OFFLOAD_BATCH_FK = : B2
 AND MMT.MONEY_MOVEMENT_TRANSACTION_SEQ = FT.MONEY_MOVEMENT_TRANSACTION_FK
 AND FT.CURRENT_TRANSACTION_STATE_FK = 'Executed'
 AND MMT.INITIATION_DATE = : B1
 AND FT.SETTLEMENT_DATE >= MMT.INITIATION_DATE
 AND TRUNC(FT.SETTLEMENT_DATE) >= TRUNC(MMT.INITIATION_DATE)
)
```

Plan hash value: 1307286917

| -----   |                                           |        |       |                                |      |       |          |
|---------|-------------------------------------------|--------|-------|--------------------------------|------|-------|----------|
| Id      | Operation                                 |        |       | Name                           | Rows | Bytes | Cost (%) |
| CPU)    | Time                                      | Pstart | Pstop |                                |      |       |          |
| -----   |                                           |        |       |                                |      |       |          |
| 0       | INSERT STATEMENT                          |        |       |                                |      |       | 883      |
| (100)   |                                           |        |       |                                |      |       |          |
| 1       | LOAD TABLE CONVENTIONAL                   |        |       | PSP_FINANCIAL_TRANS_STATE      |      |       |          |
|         |                                           |        |       |                                |      |       |          |
| 2       | NESTED LOOPS                              |        |       |                                | 1    | 234   |          |
| 883 (0) | 00:00:01                                  |        |       |                                |      |       |          |
| 3       | NESTED LOOPS                              |        |       |                                | 13   | 234   |          |
| 883 (0) | 00:00:01                                  |        |       |                                |      |       |          |
| 4       | PARTITION RANGE SINGLE                    |        |       |                                | 13   | 1105  |          |
| 506 (0) | 00:00:01                                  | KEY    | KEY   |                                |      |       |          |
| * 5     | TABLE ACCESS BY LOCAL INDEX ROWID BATCHED |        |       | PSP_MONEY_MOVEMENT_TRANSACTION | 13   | 1105  |          |
| 506 (0) | 00:00:01                                  | KEY    | KEY   |                                |      |       |          |
| * 6     | INDEX RANGE SCAN                          |        |       | PSP_MONEY_MOVEMENT_TRANSAC_FK3 | 5216 |       |          |

```

58 (0)| 00:00:01 | KEY | KEY |
| 7 | PARTITION RANGE AND | 1 |
27 (0)| 00:00:01 |KEY(AP)| 82 |
|* 8 | INDEX RANGE SCAN | PSP_FINANCIAL_TRANSACTION_FK10 | 1 |
27 (0)| 00:00:01 |KEY(AP)| 82 |
|* 9 | TABLE ACCESS BY LOCAL INDEX ROWID | PSP_FINANCIAL_TRANSACTION | 1 | 149 |
29 (0)| 00:00:01 | 1 | 1 |

```

Predicate Information (identified by operation id):

```

5 - filter("MMT"."INITIATION_DATE"=:B1)
6 - access("MMT"."OFFLOAD_BATCH_FK"=:B2)
8 - access("MMT"."MONEY_MOVEMENT_TRANSACTION_SEQ"="FT"."MONEY_MOVEMENT_TRANSACTION_FK")
9 - filter(("FT"."SETTLEMENT_DATE">=:B1 AND "FT"."CURRENT_TRANSACTION_STATE_FK"='Executed' AND
TRUNC(INTERNAL_FUNCTION("FT"."SETTLEMENT_DATE"))>=TRUNC(INTERNAL_FUNCTION("MMT"."
INITIATION_DATE"))) AND
"FT"."SETTLEMENT_DATE">="MMT"."INITIATION_DATE"))

```

Note

- Direct Load disabled because parent referential constraints are present

## Opportunity #17

### DailyBatchJobsProcessor - fx6bn8avj7cp2

```
BEGIN PRC_OFFLOAD_INSERT_FTS(:1 , :2 , :3 , :4); END;
```

**Note:** Opportunity #16 sql is part of this procedure.

SQL ordered by Reads (21-Sep-21 19:00 - 21-Sep-21 19:45)

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                    | SQL Text                             |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|-------------------------------|--------------------------------------|
| 11,143,921     | 1          | 11,143,921.00  | 52.68   | 956.35           | 72.54 | 28.14 | 6qk524uazdf0c | GemsAccountsReceivableProcess | UPDATE<br>PSP_FINANCIAL_TRANS_STA... |

SQL ordered by Reads (21-Sep-21 18:30 - 21-Sep-21 19:00)

| Physical Reads | Executions | Reads per Exec | %Total | Elapsed Time (s) | %CPU  | %IO   | SQL Id        | SQL Module                    | SQL Text                          |
|----------------|------------|----------------|--------|------------------|-------|-------|---------------|-------------------------------|-----------------------------------|
| 6,617,455      | 1          | 6,617,455.00   | 5.95   | 1,110.69         | 15.59 | 89.10 | 5nmd8a96rcf9j | BaseATFExtractFileProcess:146 | /* Select ptp.Law , ptp.Amount... |
| 4,713,495      | 1          | 4,713,495.00   | 4.24   | 106.27           | 14.73 | 84.23 | fud06g9jzyy2  | BaseATFExtractFileProcess:146 | /* Select companyDailyLiabilit... |
| 2,501,012      | 1          | 2,501,012.00   | 2.25   | 200.03           | 7.91  | 93.61 | 5tnr2d161kga  | BaseATFExtractFileProcess:146 | /* select co, serviceInfo.Stat... |
| 1,269,865      | 1          | 1,269,865.00   | 1.14   | 153.67           | 6.96  | 94.24 | a1zfnguds0wgx | BaseATFExtractFileProcess:146 | /* select comp.SourceCompanyId... |

SQL ordered by Reads (20-Sep-21 19:00 - 20-Sep-21 19:45)

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                    | SQL Text                             |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|-------------------------------|--------------------------------------|
| 8,791,805      | 1          | 8,791,805.00   | 52.09   | 898.46           | 71.21 | 29.66 | 6qk524uazdf0c | GemsAccountsReceivableProcess | UPDATE<br>PSP_FINANCIAL_TRANS_STA... |

SQL ordered by Reads (20-Sep-21 18:30 - 20-Sep-21 19:00)

| Physical Reads | Executions | Reads per Exec | % Total | Elapsed Time (s) | % CPU | %IO   | SQL Id        | SQL Module                           | SQL Text                             |
|----------------|------------|----------------|---------|------------------|-------|-------|---------------|--------------------------------------|--------------------------------------|
| 76,127,056     | 0          |                | 70.15   | 592.00           | 50.35 | 50.60 | 6qk524uazdf0c | GemsAccountsReceivableProcesso<br>r: | UPDATE<br>PSP_FINANCIAL_TRANS_STA... |
| 4,713,495      | 1          | 4,713,495.00   | 4.34    | 99.85            | 17.02 | 25.37 | fud06g9jzyyh2 | BaseATFExtractFileProcess:146        | /* Select companyDailyLiabilit...    |
| 3,415,538      | 1          | 3,415,538.00   | 3.15    | 192.68           | 10.64 | 70.72 | 5nmd8a96rcf9j | BaseATFExtractFileProcess:146        | /* Select ptp.Law , ptp.Amount...    |
| 2,515,997      | 1          | 2,515,997.00   | 2.32    | 212.07           | 7.74  | 93.06 | 5ttnr2d161kga | BaseATFExtractFileProcess:146        | /* select co, serviceInfo.Stat...    |
| 1,271,218      | 1          | 1,271,218.00   | 1.17    | 160.79           | 6.87  | 87.16 | a1zfnguds0wgx | BaseATFExtractFileProcess:146        | /* select comp.SourceCompanyId...    |

## More Queries to check

### BatchJob (Module: BaseATFExtractFileProcess)

sql\_id: f8ct99p7ryv1d

```

/* dynamic native SQL query */ select opc.source_company_id,
 ope.source_employee_id,
 opwp.wage_plan_domain,
 opwp.state,
 opwp.name,
 opwp.wage_plan_value,
 opwp.description
from psp_employee ope
 join psp_employee_wage_plan opwp on ope.employee_seq = opwp.employee_fk
 join psp_company opc on ope.COMPANY_FK = opc.COMPANY_SEQ
where opwp.invalid_date is null
and opc.company_seq in (
 select distinct pc.company_seq
 from psp_employee pe
 join psp_employee_wage_plan pwp on pe.employee_seq = pwp.employee_fk
 join psp_company pc on PE.COMPANY_FK = PC.COMPANY_SEQ
 join psp_company_service pcs on PCS.COMPANY_FK = PC.COMPANY_SEQ
 join psp_tax_company_service_info ptcsinfo
 on PTCSINFO.TAX_COMPANY_SERVICE_INFO_SEQ = PCS.
COMPANY_SERVICE_SEQ
 join psp_qbdt_employee_info qei on pe.employee_seq = qei.employee_fk
where qei.is_assisted = 1
and ((PCS.STATUS_CD = 'ActiveCurrent' and PCS.SERVICE_START_DATE is not
null)
 or (PCS.STATUS_CD in ('Cancelled', 'Terminated') and
 PTCSINFO.LAST_QUARTER_TO_FILE <> 0))
and pwp.invalid_date is null)
order by opc.SOURCE_COMPANY_ID, ope.SOURCE_EMPLOYEE_ID, opwp.
wage_plan_value

```

| I  | SQL_PROFILE<br>(LAST_ACTIVE_T | SQL_PLAN_BASELI<br>LIO | SQL_ID<br>PIO | CHILD_NUMBER<br>AVG_CPU_TIME_PER_EXEC | PLAN_HASH_VALUE<br>IO_TIME | EXECUTIONS<br>AVG_ROWS | EL_PER_EXEC<br>FIRST_LOAD_TIME | TO_CHAR    |
|----|-------------------------------|------------------------|---------------|---------------------------------------|----------------------------|------------------------|--------------------------------|------------|
| N  |                               |                        | f8ct99p7ryv1d | 0                                     | 2574391374                 | 1                      | 26.902583                      | 25-DEC 05: |
| 08 | 14739822                      | 73580                  | 23.5462       | 3.2474                                | 663677                     | 2021-12-25/05:06:47    |                                |            |
| Y  |                               |                        | f8ct99p7ryv1d | 1                                     | 3358292017                 | 1                      | 688.814935                     | 26-DEC 05: |
| 40 | 696870213                     | 73583                  | 683.2168      | 5.7521                                | 663680                     | 2021-12-25/05:06:47    |                                |            |

plan history:

| SQL_PROFILE    | SNAP_ID | BEGIN_INTERVAL_TIME       | SQL_ID        | PLAN_HASH_VALUE | EXECS | ELAPSED_PER_EXEC | CPU_PER_EXEC |
|----------------|---------|---------------------------|---------------|-----------------|-------|------------------|--------------|
| GETS_PER_EXEC  | 36711   | 05-OCT-21 03.45.12.778 AM | f8ct99p7ryv1d | 2574391374      | 1     | 60.709           |              |
| READS_PER_EXEC | 26.303  | 13421500                  | 135309        | 654029          |       |                  |              |
| ROWS_PER_EXEC  | 40947   | 18-NOV-21 06.45.12.243 AM | f8ct99p7ryv1d | 2574391374      | 1     | 176.706          |              |
|                | 36.034  | 13591656                  | 369621        | 655883          |       |                  |              |
|                | 43801   | 17-DEC-21 11.15.29.121 PM | f8ct99p7ryv1d | 2574391374      | 1     | 28.895           |              |
|                | 25.376  | 14729645                  | 72441         | 663062          |       |                  |              |
|                | 44496   | 25-DEC-21 05.00.22.478 AM | f8ct99p7ryv1d | 2574391374      | 1     | 26.903           |              |
|                | 23.546  | 14739822                  | 73580         | 663677          |       |                  |              |

4 rows selected.

plan

Plan hash value: 3358292017

| Id   | Operation                           | Name                         | Rows  | Bytes | TempSpc | Cost (%CPU) | Time     |
|------|-------------------------------------|------------------------------|-------|-------|---------|-------------|----------|
| 0    | SELECT STATEMENT                    |                              |       |       |         | 193K (100)  |          |
| 1    | SORT ORDER BY                       |                              | 25292 | 36M   |         | 193K (1)    | 00:00:08 |
| 2    | VIEW                                | VM_NWVW_2                    | 25292 | 36M   |         | 193K (1)    | 00:00:08 |
| 3    | HASH UNIQUE                         |                              | 25292 | 14M   |         | 193K (1)    | 00:00:08 |
| 4    | NESTED LOOPS                        |                              | 6     | 3492  |         | 193K (1)    | 00:00:08 |
| 5    | NESTED LOOPS                        |                              | 504   | 233K  |         | 192K (1)    | 00:00:08 |
| 6    | NESTED LOOPS                        |                              | 58    | 22272 |         | 191K (1)    | 00:00:08 |
| 7    | NESTED LOOPS SEMI                   |                              | 57    | 18525 |         | 191K (1)    | 00:00:08 |
| * 8  | HASH JOIN SEMI                      |                              | 499   | 138K  | 12M     | 189K (1)    | 00:00:08 |
| 9    | NESTED LOOPS                        |                              | 51442 | 11M   |         | 161K (1)    | 00:00:07 |
| 10   | NESTED LOOPS                        |                              | 5891  | 977K  |         | 116K (1)    | 00:00:05 |
| * 11 | HASH JOIN                           |                              | 5781  | 750K  | 7048K   | 104K (1)    | 00:00:05 |
| 12   | TABLE ACCESS FULL                   | PSP_TAX_COMPANY_SERVICE_INFO | 136K  | 5450K |         | 488 (1)     | 00:00:01 |
| 13   | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE          | 8915K | 782M  |         | 60034 (1)   | 00:00:03 |
| * 14 | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     | 37    |         | 2 (0)       | 00:00:01 |
| 15   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 666   |         | 18 (0)      | 00:00:01 |
| * 16 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         | 3 (0)       | 00:00:01 |
| * 17 | TABLE ACCESS FULL                   | PSP_EMPLOYEE_WAGE_PLAN       | 712K  | 27M   |         | 24917 (1)   | 00:00:01 |
| * 18 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_EMPLOYEE_INFO       | 763K  | 29M   |         | 5 (0)       | 00:00:01 |
| * 19 | INDEX RANGE SCAN                    | PSP_QBDT_EMPLOYEE_INFO_FK1   | 1     |       |         | 3 (0)       | 00:00:01 |
| 20   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1     | 59    |         | 4 (0)       | 00:00:01 |
| * 21 | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     |       |         | 2 (0)       | 00:00:01 |
| 22   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 819   |         | 18 (0)      | 00:00:01 |
| * 23 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         | 3 (0)       | 00:00:01 |
| * 24 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE_WAGE_PLAN       | 1     | 107   |         | 5 (0)       | 00:00:01 |
| * 25 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_WAGE_PLAN_FK1   | 1     |       |         | 3 (0)       | 00:00:01 |

Predicate Information (identified by operation id):

```
8 - access("PE"."EMPLOYEE_SEQ"="PWP"."EMPLOYEE_FK")
11 - access("PTCSINFO"."TAX_COMPANY_SERVICE_INFO_SEQ"="PCS"."COMPANY_SERVICE_SEQ")
 filter((((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
 (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSINFO"."LAST_QUARTER_TO_FILE"<>0))))
14 - access("PCS"."COMPANY_FK"="PC"."COMPANY_SEQ")
16 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
17 - filter("PWP"."INVALID_DATE" IS NULL)
18 - filter("QEI"."IS_ASSISTED"=1)
19 - access("PE"."EMPLOYEE_SEQ"="QEI"."EMPLOYEE_FK")
21 - access("OPC"."COMPANY_SEQ"="PC"."COMPANY_SEQ")
23 - access("OPE"."COMPANY_FK"="OPC"."COMPANY_SEQ")
24 - filter("OPWP"."INVALID_DATE" IS NULL)
25 - access("OPE"."EMPLOYEE_SEQ"="OPWP"."EMPLOYEE_FK")
```

Note

- statistics feedback used for this statement
- this is an adaptive plan

Plan hash value: 2574391374

| Id   | Operation                           | Name                         | Rows  | Bytes | TempSpc | Cost (%CPU) | Time     |
|------|-------------------------------------|------------------------------|-------|-------|---------|-------------|----------|
| 0    | SELECT STATEMENT                    |                              |       |       |         | 193K(100)   |          |
| 1    | SORT ORDER BY                       |                              | 6     | 1548  |         | 193K (1)    | 00:00:08 |
| 2    | NESTED LOOPS                        |                              | 6     | 1548  |         | 193K (1)    | 00:00:08 |
| 3    | NESTED LOOPS                        |                              | 507   | 1548  |         | 193K (1)    | 00:00:08 |
| 4    | NESTED LOOPS                        |                              | 507   | 82641 |         | 192K (1)    | 00:00:08 |
| 5    | NESTED LOOPS                        |                              | 58    | 4872  |         | 191K (1)    | 00:00:08 |
| 6    | VIEW                                | VW_NSQ_1                     | 57    | 2109  |         | 191K (1)    | 00:00:08 |
| 7    | HASH UNIQUE                         |                              | 57    | 18525 |         | 191K (1)    | 00:00:08 |
| 8    | NESTED LOOPS SEMI                   |                              | 57    | 18525 |         | 191K (1)    | 00:00:08 |
| * 9  | HASH JOIN SEMI                      |                              | 499   | 138K  | 12M     | 189K (1)    | 00:00:08 |
| 10   | NESTED LOOPS                        |                              | 51442 | 11M   |         | 161K (1)    | 00:00:07 |
| 11   | NESTED LOOPS                        |                              | 5891  | 977K  |         | 116K (1)    | 00:00:05 |
| * 12 | HASH JOIN                           |                              | 5781  | 750K  | 7048K   | 104K (1)    | 00:00:05 |
| 13   | TABLE ACCESS FULL                   | PSP_TAX_COMPANY_SERVICE_INFO | 136K  | 5450K |         | 488 (1)     | 00:00:01 |
| 14   | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE          | 8915K | 782M  |         | 60034 (1)   | 00:00:03 |
| * 15 | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     | 37    |         | 2 (0)       | 00:00:01 |
| 16   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 666   |         | 18 (0)      | 00:00:01 |
| * 17 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         | 3 (0)       | 00:00:01 |
| * 18 | TABLE ACCESS FULL                   | PSP_EMPLOYEE_WAGE_PLAN       | 712K  | 27M   |         | 24917 (1)   | 00:00:01 |
| * 19 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_EMPLOYEE_INFO       | 763K  | 29M   |         | 5 (0)       | 00:00:01 |
| * 20 | INDEX RANGE SCAN                    | PSP_QBDT_EMPLOYEE_INFO_FK1   | 1     |       |         | 3 (0)       | 00:00:01 |
| 21   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1     | 47    |         | 4 (0)       | 00:00:01 |
| * 22 | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     |       |         | 2 (0)       | 00:00:01 |
| 23   | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 711   |         | 18 (0)      | 00:00:01 |
| * 24 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         | 3 (0)       | 00:00:01 |
| * 25 | INDEX RANGE SCAN                    | PSP_EMPLOYEE_WAGE_PLAN_FK1   | 1     |       |         | 3 (0)       | 00:00:01 |
| * 26 | TABLE ACCESS BY INDEX ROWID         | PSP_EMPLOYEE_WAGE_PLAN       | 1     | 95    |         | 5 (0)       | 00:00:01 |

Predicate Information (identified by operation id):

```
9 - access("PE"."EMPLOYEE_SEQ"="PWP"."EMPLOYEE_FK")
12 - access("PTCSINFO"."TAX_COMPANY_SERVICE_INFO_SEQ"="PCS"."COMPANY_SERVICE_SEQ")
 filter((((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
 (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSINFO"."LAST_QUARTER_TO_FILE"<>0)))
15 - access("PCS"."COMPANY_FK"="PC"."COMPANY_SEQ")
17 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
18 - filter("PWP"."INVALID_DATE" IS NULL)
19 - filter("QEI"."IS_ASSISTED"=1)
20 - access("PE"."EMPLOYEE_SEQ"="QEI"."EMPLOYEE_FK")
22 - access("OPC"."COMPANY_SEQ"="COMPANY_SEQ")
24 - access("OPE"."COMPANY_FK"="OPC"."COMPANY_SEQ")
25 - access("OPE"."EMPLOYEE_SEQ"="OPWP"."EMPLOYEE_FK")
26 - filter("OPWP"."INVALID_DATE" IS NULL)
```

Note

-----  
- this is an adaptive plan

## BatchJob (Module: BaseATFExtractFileProcess)

sql\_id: 700vqjcjvmtva

```
/* dynamic native SQL query */ select PC.SOURCE_COMPANY_ID,
PE.SOURCE_EMPLOYEE_ID,
PE.TAX_ID_ENC,
PI.FIRST_NAME,
PI.MIDDLE_NAME,
PI.LAST_NAME,
PA.ADDRESS_LINE1,
PA.ADDRESS_LINE2,
PA.CITY,
PA.STATE,
PA.ZIP_CODE,
PA.ZIP_CODE_EXTENSION,
PI.GENDER_CD,
PE.HIRE_DATE,
PE.RE_HIRE_DATE,
PE.TERMINATION_DATE,
PE.LIVE_STATE,
```

```

 PE.WORK_STATE,
 PI.PHONE,
 PE.IS_STATUTORY,
 PE.HAS_RETIREMENT_PLAN,
 PE.STATUS_CD,
 pe.fed_allowances,
 qei.EMPLOYEE_SEASONAL
 from psp_employee pe
 join psp_company pc on PE.COMPANY_FK = PC.COMPANY_SEQ
 join psp_company_service pcs on PCS.COMPANY_FK = PC.COMPANY_SEQ
 join psp_tax_company_service_info ptcsinfo
 on PTCSINFO.TAX_COMPANY_SERVICE_INFO_SEQ = PCS.COMPANY_SERVICE_SEQ
 join psp_individual pi on PI.INDIVIDUAL_SEQ = PE.EMPLOYEE_SEQ
 join psp_address pa on PA.ADDRESS_SEQ = PI.MAILING_ADDRESS_FK
 join psp_qbdt_employee_info qei on pe.employee_seq = qei.employee_fk
 where qei.is_assisted = 1
 and ((PCS.STATUS_CD = 'ActiveCurrent' and PCS.SERVICE_START_DATE is not null)
 or (PCS.STATUS_CD in ('Cancelled', 'Terminated') and
 PTCSINFO.LAST_QUARTER_TO_FILE <> 0))
 order by PC.SOURCE_COMPANY_ID, PE.SOURCE_EMPLOYEE_ID

```

| I SQL_PROFILE                     | SQL_PLAN_BASELI | SQL_ID        | CHILD_NUMBER          | PLAN_HASH_VALUE | EXECUTIONS         |
|-----------------------------------|-----------------|---------------|-----------------------|-----------------|--------------------|
| EL_PER_EXEC TO_CHAR(LAST_ACTIVE_T | LIO             | PIO           | AVG_CPU_TIME_PER_EXEC | IO_TIME         | AVG_ROWS           |
| FIRST_LOAD_TIME                   |                 |               |                       |                 |                    |
| N                                 |                 | 700vqjcjvmtva | 0                     | 72008934        | 1                  |
| 1731.53434 25-DEC 05:06           | 77075873        | 3863666       | 220.6181              | 1560.4331       | 6629504 2021-12-25 |
| /04:15:07                         |                 |               |                       |                 |                    |
| Y                                 |                 | 700vqjcjvmtva | 1                     | 72008934        | 1                  |
| 2175.22871 26-DEC 05:27           | 77076445        | 4909210       | 243.4219              | 1992.238        | 6629529 2021-12-25 |
| /04:15:07                         |                 |               |                       |                 |                    |

Plan hash value: 72008934

| Id        | Operation                           | Name                         | Rows  | Bytes | TempSpc | Cost |
|-----------|-------------------------------------|------------------------------|-------|-------|---------|------|
| (%CPU)    | Time                                |                              |       |       |         |      |
| 0         | SELECT STATEMENT                    |                              |       |       |         |      |
| 459K(100) |                                     |                              |       |       |         |      |
| 1         | SORT ORDER BY                       |                              | 4901  | 2675K | 2816K   |      |
| 459K (1)  | 00:00:18                            |                              |       |       |         |      |
| 2         | NESTED LOOPS                        |                              | 4901  | 2675K |         |      |
| 459K (1)  | 00:00:18                            |                              |       |       |         |      |
| 3         | NESTED LOOPS                        |                              | 4868  | 2329K |         |      |
| 434K (1)  | 00:00:17                            |                              |       |       |         |      |
| 4         | NESTED LOOPS                        |                              | 5834  | 2284K |         |      |
| 411K (1)  | 00:00:17                            |                              |       |       |         |      |
| 5         | NESTED LOOPS                        |                              | 51442 | 17M   |         |      |
| 173K (1)  | 00:00:07                            |                              |       |       |         |      |
| 6         | NESTED LOOPS                        |                              | 5891  | 1035K |         |      |
| 127K (1)  | 00:00:05                            |                              |       |       |         |      |
| * 7       | HASH JOIN                           |                              | 5781  | 750K  | 7048K   |      |
| 104K (1)  | 00:00:05                            |                              |       |       |         |      |
| 8         | TABLE ACCESS FULL                   | PSP_TAX_COMPANY_SERVICE_INFO | 136K  | 5450K |         |      |
| 488 (1)   | 00:00:01                            |                              |       |       |         |      |
| 9         | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE          | 8915K | 782M  |         |      |
| 60034 (1) | 00:00:03                            |                              |       |       |         |      |
| 10        | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1     | 47    |         |      |
| 4 (0)     | 00:00:01                            |                              |       |       |         |      |
| * 11      | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     |       |         |      |
| 2 (0)     | 00:00:01                            |                              |       |       |         |      |
| 12        | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 1602  |         |      |
| 18 (0)    | 00:00:01                            |                              |       |       |         |      |
| * 13      | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         |      |
| 3 (0)     | 00:00:01                            |                              |       |       |         |      |
| * 14      | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_QBDT_EMPLOYEE_INFO       | 1     | 43    |         |      |



```

5 (0)| 00:00:01 |
|* 15 | INDEX RANGE SCAN | PSP_QBDT_EMPLOYEE_INFO_FK1 | 1 | | |
3 (0)| 00:00:01 |
|* 16 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_INDIVIDUAL | 1 | 89 | |
4 (0)| 00:00:01 |
|* 17 | INDEX RANGE SCAN | SYS_C0020489 | 1 | | |
3 (0)| 00:00:01 |
| 18 | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_ADDRESS | 1 | 69 | |
5 (0)| 00:00:01 |
|* 19 | INDEX RANGE SCAN | SYS_C0020476 | 1 | | |
3 (0)| 00:00:01 |

```

Predicate Information (identified by operation id):

```

7 - access("PTCSINFO"."TAX_COMPANY_SERVICE_INFO_SEQ"="PCS"."COMPANY_SERVICE_SEQ")
 filter(((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
 (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSINFO"."LAST_QUARTER_TO_FILE"<>0))))
11 - access("PCS"."COMPANY_FK"="PC"."COMPANY_SEQ")
13 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
14 - filter("QEI"."IS_ASSISTED"=1)
15 - access("PE"."EMPLOYEE_SEQ"="QEI"."EMPLOYEE_FK")
16 - filter("PI"."MAILING_ADDRESS_FK" IS NOT NULL)
17 - access("PI"."INDIVIDUAL_SEQ"="PE"."EMPLOYEE_SEQ")
19 - access("PA"."ADDRESS_SEQ"="PI"."MAILING_ADDRESS_FK")

```

Note

- this is an adaptive plan

## BatchJob (Module: BaseATFExtractFileProcess)

sql\_id: 700vqjcjvmtva

```

/* dynamic native SQL query */
SELECT PC.SOURCE_COMPANY_ID AS PSID,
 PE.SOURCE_EMPLOYEE_ID AS EMPLOYEE_ID,
 PCL.LAW_FK AS LAW_ID,
 MD.VALUE AS VALUE
FROM PSP_TAX_TABLE_MISC_DATA MD
 JOIN PSP_EMPLOYEE_TAX PET ON MD.EMPLOYEE_TAX_FK = PET.EMPLOYEE_TAX_SEQ
 JOIN PSP_COMPANY_LAW PCL ON PET.COMPANY_LAW_FK = PCL.COMPANY_LAW_SEQ
 JOIN PSP_EMPLOYEE PE ON PET.EMPLOYEE_FK = PE.EMPLOYEE_SEQ
 JOIN PSP_COMPANY PC ON PE.COMPANY_FK = PC.COMPANY_SEQ
 JOIN PSP_COMPANY_SERVICE PCS ON PC.COMPANY_SEQ = PCS.COMPANY_FK
 JOIN PSP_TAX_COMPANY_SERVICE_INFO PTCSI
 ON PCS.COMPANY_SERVICE_SEQ = PTCSI.TAX_COMPANY_SERVICE_INFO_SEQ
WHERE MD.VALUE IS NOT NULL
 AND ((PCS.STATUS_CD = 'ActiveCurrent' AND PCS.SERVICE_START_DATE IS NOT NULL)
 OR (PCS.STATUS_CD IN ('Cancelled', 'Terminated') AND
 PTCSI.LAST_QUARTER_TO_FILE <> 0))
order by PC.SOURCE_COMPANY_ID, PE.SOURCE_EMPLOYEE_ID;

I SQL_PROFILE SQL_PLAN_BASELI SQL_ID CHILD_NUMBER PLAN_HASH_VALUE EXECUTIONS
EL_PER_EXEC TO_CHAR(LAST_ACTIVE_T LIO PIO AVG_CPU_TIME_PER_EXEC IO_TIME AVG_ROWS
FIRST_LOAD_TIME

N 7pwsv4u6w1tj6 0 2815818285 1
3880.08438 25-DEC 06:15 78175420 9372167 367.5081 3632.6517 2099660 2021-12-25
/05:08:02
Y 7pwsv4u6w1tj6 1 2815818285 1
4601.41689 26-DEC 06:58 78316447 10892924 431.4044 4331.9265 2099676 2021-12-25
/05:08:02

```

Plan hash value: 2815818285

| Id   Operation |                                     | Name                         | Rows  | Bytes | TempSpc | Cost |
|----------------|-------------------------------------|------------------------------|-------|-------|---------|------|
| (%CPU)         | Time                                |                              |       |       |         |      |
| 0              | SELECT STATEMENT                    |                              |       |       |         |      |
| 550K(100)      |                                     |                              |       |       |         |      |
| 1              | SORT ORDER BY                       |                              | 20053 | 8244K | 8928K   |      |
| 550K (1)       | 00:00:22                            |                              |       |       |         |      |
| 2              | NESTED LOOPS                        |                              | 20053 | 8244K |         |      |
| 548K (1)       | 00:00:22                            |                              |       |       |         |      |
| 3              | NESTED LOOPS                        |                              | 19898 | 7384K |         |      |
| 468K (1)       | 00:00:19                            |                              |       |       |         |      |
| 4              | NESTED LOOPS                        |                              | 27675 | 9215K |         |      |
| 361K (1)       | 00:00:15                            |                              |       |       |         |      |
| 5              | NESTED LOOPS                        |                              | 51442 | 12M   |         |      |
| 173K (1)       | 00:00:07                            |                              |       |       |         |      |
| 6              | NESTED LOOPS                        |                              | 5891  | 1035K |         |      |
| 127K (1)       | 00:00:05                            |                              |       |       |         |      |
| * 7            | HASH JOIN                           |                              | 5781  | 750K  | 7048K   |      |
| 104K (1)       | 00:00:05                            |                              |       |       |         |      |
| 8              | TABLE ACCESS FULL                   | PSP_TAX_COMPANY_SERVICE_INFO | 136K  | 5450K |         |      |
| 488 (1)        | 00:00:01                            |                              |       |       |         |      |
| 9              | TABLE ACCESS FULL                   | PSP_COMPANY_SERVICE          | 8915K | 782M  |         |      |
| 60034 (1)      | 00:00:03                            |                              |       |       |         |      |
| 10             | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY                  | 1     | 47    |         |      |
| 4 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 11           | INDEX RANGE SCAN                    | SYS_C0020424                 | 1     |       |         |      |
| 2 (0)          | 00:00:01                            |                              |       |       |         |      |
| 12             | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE                 | 9     | 711   |         |      |
| 18 (0)         | 00:00:01                            |                              |       |       |         |      |
| * 13           | INDEX RANGE SCAN                    | PSP_EMPLOYEE_FK2             | 28    |       |         |      |
| 3 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 14           | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_EMPLOYEE_TAX             | 1     | 82    |         |      |
| 5 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 15           | INDEX RANGE SCAN                    | PSP_EMPLOYEE_TAX_FK2         | 9     |       |         |      |
| 3 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 16           | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_TAX_TABLE_MISC_DATA      | 1     | 39    |         |      |
| 5 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 17           | INDEX RANGE SCAN                    | PSP_TAX_TABLE_MISC_DATA_FK1  | 3     |       |         |      |
| 3 (0)          | 00:00:01                            |                              |       |       |         |      |
| 18             | TABLE ACCESS BY INDEX ROWID BATCHED | PSP_COMPANY_LAW              | 1     | 41    |         |      |
| 4 (0)          | 00:00:01                            |                              |       |       |         |      |
| * 19           | INDEX RANGE SCAN                    | SYS_C0020562                 | 1     |       |         |      |
| 2 (0)          | 00:00:01                            |                              |       |       |         |      |

Predicate Information (identified by operation id):

- ```
7 - access("PCS"."COMPANY_SERVICE_SEQ"="PTCSI"."TAX_COMPANY_SERVICE_INFO_SEQ")
    filter(((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
              (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSI"."LAST_QUARTER_TO_FILE"<>0)))
11 - access("PC"."COMPANY_SEQ"="PCS"."COMPANY_FK")
13 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
14 - filter("PET"."COMPANY_LAW_FK" IS NOT NULL)
15 - access("PET"."EMPLOYEE_FK"="PE"."EMPLOYEE_SEQ")
16 - filter("MD"."VALUE" IS NOT NULL)
17 - access("MD"."EMPLOYEE_TAX_FK"="PET"."EMPLOYEE_TAX_SEQ")
19 - access("PET"."COMPANY_LAW_FK"="PCL"."COMPANY_LAW_SEQ")
```

Note

- this is an adaptive plan

FYI, below queries got changed and no more opportunity for optimization.

1. BaseATFExtractFileProcess - 36v8ajy4fr4z

Discontinued after 17-Dec-2021

```
/* dynamic native SQL query */
select
  PC.SOURCE_COMPANY_ID,
  PE.SOURCE_EMPLOYEE_ID,
  PE.TAX_ID_ENC,
  PI.FIRST_NAME,
  PI.MIDDLE_NAME,
  PI.LAST_NAME,
  PA.ADDRESS_LINE1,
  PA.ADDRESS_LINE2,
  PA.CITY,
  PA.STATE,
  PA.ZIP_CODE,
  PA.ZIP_CODE_EXTENSION,
  PI.GENDER_CD,
  PE.HIRE_DATE,
  PE.RE_HIRE_DATE,
  PE.TERMINATION_DATE,
  PE.LIVE_STATE,
  PE.WORK_STATE,
  PI.PHONE,
  PE.IS_STATUTORY,
  PE.HAS_RETIREMENT_PLAN,
  PE.STATUS_CD,
  pe.fed_allowances,
```

```

    qei.EMPLOYEE_SEASONAL
from
    psp_employee pe
        join psp_company pc on PE.COMPANY_FK = PC.COMPANY_SEQ
        join psp_company_service pcs on PCS.COMPANY_FK = PC.COMPANY_SEQ
        join psp_tax_company_service_info ptcsinfo on PTCSINFO.TAX_COMPANY_SERVICE_INFO_SEQ = PCS.
COMPANY_SERVICE_SEQ
        join psp_individual pi on PI.INDIVIDUAL_SEQ = PE.EMPLOYEE_SEQ
    join psp_address pa on PA.ADDRESS_SEQ = PI.MAILING_ADDRESS_FK
    join psp_qbdt_employee_info qei on pe.employee_seq = qei.employee_fk
where
    qei.is_assisted = 1
    and (
        (
            PCS.STATUS_CD = 'ActiveCurrent'
        and PCS.SERVICE_START_DATE is not null
        )
        or (
            PCS.STATUS_CD in ('Cancelled', 'Terminated')
        and PTCSINFO.LAST_QUARTER_TO_FILE <> 0
        )
    )
    and (
        (
            PI.MODIFIED_DATE >= : 1
        and PI.MODIFIED_DATE < : 2
        )
        or (
            PA.MODIFIED_DATE >= : 3
        and PA.MODIFIED_DATE < : 4
        )
        or PCS.STATUS_EFFECTIVE_DATE >= : 5
    )
    and PC.IS_DG_DISASSOCIATED = 0
order by
    PC.SOURCE_COMPANY_ID,
    PE.SOURCE_EMPLOYEE_ID

```

Plan hash value: 72008934

Id		Operation	Name	Rows	Bytes	TempSpc	Cost
(%CPU)	Time						
0		SELECT STATEMENT					
458K(100)							
1		SORT ORDER BY		1	594		
458K (1)	00:00:18						
2		NESTED LOOPS		1	594		
458K (1)	00:00:18						
3		NESTED LOOPS		4820	2419K		
434K (1)	00:00:17						
4		NESTED LOOPS		5796	2348K		
411K (1)	00:00:17						
5		NESTED LOOPS		50553	17M		
177K (1)	00:00:07						
6		NESTED LOOPS		5888	1115K		
132K (1)	00:00:06						
7		HASH JOIN		5781	812K	7048K	
109K (1)	00:00:05						
8		TABLE ACCESS FULL	PSP_TAX_COMPANY_SERVICE_INFO	136K	5450K		
488 (1)	00:00:01						
9		TABLE ACCESS FULL	PSP_COMPANY_SERVICE	8915K	875M		
60034 (1)	00:00:03						
10		TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY	1	50		
4 (0)	00:00:01						
11		INDEX RANGE SCAN	SYS_C0020424	1			
2 (0)	00:00:01						

12	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE	9	1602	
18	(0) 00:00:01				
13	INDEX RANGE SCAN	PSP_EMPLOYEE_FK2	27		
3	(0) 00:00:01				
14	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_QBDT_EMPLOYEE_INFO	1	43	
5	(0) 00:00:01				
15	INDEX RANGE SCAN	PSP_QBDT_EMPLOYEE_INFO_FK1	1		
3	(0) 00:00:01				
16	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_INDIVIDUAL	1	99	
4	(0) 00:00:01				
17	INDEX RANGE SCAN	SYS_C0020489	1		
3	(0) 00:00:01				
18	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_ADDRESS	1	80	
5	(0) 00:00:01				
19	INDEX RANGE SCAN	SYS_C0020476	1		
3	(0) 00:00:01				

Note

- this is an adaptive plan

2. BaseATFExtractFileProcess - 3gsxm5sch6672-

Discontinued after 17-Dec-2021

```

/* dynamic native SQL query */
(
  select company.SOURCE_COMPANY_ID      as srcCompanyId,
         employee.SOURCE_EMPLOYEE_ID    as srcEmpId,
         elqt.QUARTER                    as qtr,
         elqt.YEAR                        as year,
         srcSystemLawAssoc.SOURCE_LAW_CODE as srcLawCd,
         elqt.TOTAL_WAGES                 as totalWages,
         elqt.TAXABLE_WAGES               as taxableWages,
         elqt.TIPS_TAXABLE_WAGES_AMOUNT  as tipsTaxableWageAmount,
         elqt.TAX_AMOUNT                  as taxAmount,
         CEIL(elqt.HOURS_WORKED)          as hrsWorked,
         elqt.WEEKS_WORKED                as weeksWorked,
         elqt.MONTH_ONE_WORKED_INDICATOR as m1WI,
         elqt.MONTH_TWO_WORKED_INDICATOR as m2WI,
         elqt.MONTH_THREE_WORKED_INDICATOR as m3WI,
         CEIL(elqt.HOURLY_RATE)           as hourlyRate,
         ''                               as srcPayrollItemId
  from PSP_EMPLOYEE_LAW_QTR_TOTALS elqt,
       PSP_COMPANY company,
       PSP_EMPLOYEE employee
       inner join PSP_INDIVIDUAL ind on employee.EMPLOYEE_SEQ = ind.INDIVIDUAL_SEQ,
       PSP_SOURCE_SYSTEM_LA W_ASSOC srcSystemLawAssoc,
       PSP_COMPANY_LAW companylaw
  where
    elqt.COMPANY_FK = company.COMPANY_SEQ
  and elqt.EMPLOYEE_FK = employee.EMPLOYEE_SEQ
  and elqt.COMPANY_LAW_FK = companylaw.COMPANY_LAW_SEQ
  and (
    elqt.YEAR
    >: 1
  or (
    elqt.YEAR =: 2
  and elqt.QUARTER >=: 3
    )
  )
  and (
    elqt.YEAR
    <: 4
  or (
    elqt.YEAR =: 5

```

```

and elqt.QUARTER <=: 6
)
)
and srcSystemLawAssoc.LAW_FK = elqt.LAW_FK
and srcSystemLawAssoc.SOURCE_SYSTEM_FK = 'QBDT'
and companylaw.IS_ARCHIVED = 0
and elqt.MODIFIED_DATE >=: 7
and company.IS_DG_DISASSOCIATED = 0
UNION
select
  company.SOURCE_COMPANY_ID as srcCompanyId,
  srcEmpId.SOURCE_EMPLOYEE_ID as srcEmpId,
  qet.QUARTER as qtr,
  qet.YEAR as year,
  '' as srcLawCd,
  sum(qet.TOTAL_WAGES) as totalWages,
  sum(qet.TAXABLE_WAGES) as taxableWages,
  sum(qet.TIPS_TAXABLE_WAGES_AMOUNT) as tipsTaxableWageAmount,
  sum(qet.AMOUNT) as taxAmount,
  0.0,
  0,
  0,
  0,
  0,
  0.0,
  srcPayrollItemId.SOURCE_PAYROLL_ITEM_ID as srcPayrollItemId
from
  PSP_EE_PAYROLLITEM_QTRTOTALS qet
  inner join PSP_EMPLOYEE srcEmpId
on qet.EMPLOYEE_FK = srcEmpId.EMPLOYEE_SEQ
  inner join PSP_INDIVIDUAL ind on srcEmpId.EMPLOYEE_SEQ = ind.INDIVIDUAL_SEQ
  inner join PSP_COMPANY_PAYROLL_ITEM srcPayrollItemId on qet.COMPANY_PAYROLL_ITEM_FK = srcPayrollItemId.
COMPANY_PAYROLL_ITEM_SEQ
  inner join PSP_COMPANY company on srcPayrollItemId.COMPANY_FK = company.COMPANY_SEQ
where
  (
    qet.YEAR
    >: 8
  or (
    qet.YEAR =: 9
    and qet.QUARTER >=: 10
  )
  )
  and (
    qet.YEAR
    <: 11
  or (
    qet.YEAR = : 12
    and qet.QUARTER <=: 13
  )
  )
  and qet.MODIFIED_DATE >=: 14
  and company.IS_DG_DISASSOCIATED = 0
Group by
  company.SOURCE_COMPANY_ID,
  srcEmpId.SOURCE_EMPLOYEE_ID,
  srcPayrollItemId.SOURCE_PAYROLL_ITEM_ID,
  qet.QUARTER,
  qet.YEAR
)
order by
  srcCompanyId,
  srcEmpId,
  srcLawCd,
  qtr

```

Plan hash value: 2062493046

```

-----
| Id | Operation | Name | Rows | Bytes | Cost (%CPU) |

```

Time						
0	SELECT STATEMENT					2618K(100)
1	SORT UNIQUE		2	667	2618K	(1)
00:01:43						
2	UNION-ALL					
3	NESTED LOOPS		1	358	2469K	(1)
00:01:37						
4	NESTED LOOPS SEMI		1	308	2469K	(1)
00:01:37						
* 5	HASH JOIN		1	268	2469K	(1)
00:01:37						
6	NESTED LOOPS SEMI		1	247	2469K	(1)
00:01:37						
7	NESTED LOOPS		1	210	2469K	(1)
00:01:37						
* 8	TABLE ACCESS FULL	PSP_EMPLOYEE_LAW_QTR_TOTALS	1	168	2469K	(1)
00:01:37						
9	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE	1	42	5	(0)
00:00:01						
* 10	INDEX RANGE SCAN	SYS_C0020486	1		3	(0)
00:00:01						
* 11	INDEX RANGE SCAN	SYS_C0020489	72M	2541M	3	(0)
00:00:01						
* 12	TABLE ACCESS FULL	PSP_SOURCE_SYSTEM_LAW_ASSOC	1	21	3	(0)
00:00:01						
* 13	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY_LAW	7147K	272M	4	(0)
00:00:01						
* 14	INDEX RANGE SCAN	SYS_C0020562	1		2	(0)
00:00:01						
* 15	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY	1	50	4	(0)
00:00:01						
* 16	INDEX RANGE SCAN	SYS_C0020424	1		2	(0)
00:00:01						
17	HASH GROUP BY		1	309	148K	(1)
00:00:06						
18	NESTED LOOPS		1	309	148K	(1)
00:00:06						
19	NESTED LOOPS		1	259	148K	(1)
00:00:06						
20	NESTED LOOPS		1	182	148K	(1)
00:00:06						
21	NESTED LOOPS		1	145	148K	(1)
00:00:06						
* 22	TABLE ACCESS FULL	PSP_EE_PAYROLLITEM_QTRTOTALS	1	103	148K	(1)
00:00:06						
23	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE	1	42	5	(0)
00:00:01						
* 24	INDEX RANGE SCAN	SYS_C0020486	1		3	(0)
00:00:01						
* 25	INDEX RANGE SCAN	SYS_C0020489	1	37	3	(0)
00:00:01						
26	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY_PAYROLL_ITEM	1	77	4	(0)
00:00:01						
* 27	INDEX RANGE SCAN	SYS_C0020573	1		2	(0)
00:00:01						
* 28	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY	1	50	4	(0)
00:00:01						
* 29	INDEX RANGE SCAN	SYS_C0020424	1		2	(0)
00:00:01						

Predicate Information (identified by operation id):

5 - access("SRCSYSTEMLAWASSOC"."LAW_FK"="ELQT"."LAW_FK")

```

8 - filter(("ELQT"."MODIFIED_DATE">=:7 AND ("ELQT"."YEAR">:1 OR ("ELQT"."YEAR"=:2 AND "ELQT"."QUARTER">=:3))
    AND ("ELQT"."YEAR"<:4 OR ("ELQT"."YEAR"=:5 AND "ELQT"."QUARTER"<=:6))))
10 - access("ELQT"."EMPLOYEE_FK"="EMPLOYEE"."EMPLOYEE_SEQ")
11 - access("EMPLOYEE"."EMPLOYEE_SEQ"="IND"."INDIVIDUAL_SEQ")
12 - filter("SRCSYSTEMLAWASSOC"."SOURCE_SYSTEM_FK"='QBDT')
13 - filter("COMPANYLAW"."IS_ARCHIVED"=0)
14 - access("ELQT"."COMPANY_LAW_FK"="COMPANYLAW"."COMPANY_LAW_SEQ")
15 - filter(NVL("COMPANY"."IS_DG_DISASSOCIATED",0)=0)
16 - access("ELQT"."COMPANY_FK"="COMPANY"."COMPANY_SEQ")
22 - filter(("QET"."MODIFIED_DATE">=:14 AND ("QET"."YEAR">:8 OR ("QET"."YEAR"=:9 AND "QET"."QUARTER">=:10))
AND
    ("QET"."YEAR"<:11 OR ("QET"."YEAR"=:12 AND "QET"."QUARTER"<=:13))))
24 - access("QET"."EMPLOYEE_FK"="SRCEMPID"."EMPLOYEE_SEQ")
25 - access("SRCEMPID"."EMPLOYEE_SEQ"="IND"."INDIVIDUAL_SEQ")
27 - access("QET"."COMPANY_PAYROLL_ITEM_FK"="SRCPAYROLLITEMID"."COMPANY_PAYROLL_ITEM_SEQ")
28 - filter(NVL("COMPANY"."IS_DG_DISASSOCIATED",0)=0)
29 - access("SRCPAYROLLITEMID"."COMPANY_FK"="COMPANY"."COMPANY_SEQ")

```

Note

- this is an adaptive plan

3. BaseATFExtractFileProcess - 9ta83w41jfk42. **Discontinued after 17-Dec-2021**

```

/* dynamic native SQL query */
SELECT PC.SOURCE_COMPANY_ID AS PSID,
       PE.SOURCE_EMPLOYEE_ID AS EMPLOYEE_ID,
       PCL.LAW_FK            AS LAW_ID,
       MD.VALUE              AS VALUE
FROM PSP_TAX_TABLE_MISC_DATA MD
JOIN PSP_EMPLOYEE_TAX PET ON MD.EMPLOYEE_TAX_FK = PET.EMPLOYEE_TAX_SEQ
JOIN PSP_COMPANY_LAW PCL ON PET.COMPANY_LAW_FK = PCL.COMPANY_LAW_SEQ
JOIN PSP_EMPLOYEE PE ON PET.EMPLOYEE_FK = PE.EMPLOYEE_SEQ
JOIN PSP_COMPANY PC ON PE.COMPANY_FK = PC.COMPANY_SEQ
JOIN PSP_COMPANY_SERVICE PCS ON PC.COMPANY_SEQ = PCS.COMPANY_FK
JOIN PSP_TAX_COMPANY_SERVICE_INFO PTCSI ON PCS.COMPANY_SERVICE_SEQ = PTCSI.TAX_COMPANY_SERVICE_INFO_SEQ
WHERE MD.VALUE IS NOT NULL
AND (
    (
        PCS.STATUS_CD = 'ActiveCurrent'
        AND PCS.SERVICE_START_DATE IS NOT NULL
    )
    OR (
        PCS.STATUS_CD IN ('Cancelled', 'Terminated')
        AND PTCSI.LAST_QUARTER_TO_FILE <> 0
    )
)
AND MD.MODIFIED_DATE >= : 1
and PC.IS_DG_DISASSOCIATED = 0
order by
    PC.SOURCE_COMPANY_ID,
    PE.SOURCE_EMPLOYEE_ID

```

Plan hash value: 2815818285

Id Operation		Name	Rows	Bytes	TempSpc	Cost
(%CPU)	Time					
0	SELECT STATEMENT					
459K(100)						
1	SORT ORDER BY		1	435		
459K (1)	00:00:18		1	435		
2	NESTED LOOPS		1	435		
459K (1)	00:00:18		1	394		
3	NESTED LOOPS		1	394		
459K (1)	00:00:18		1	394		

4	NESTED LOOPS		26258	8821K	
358K (1)	00:00:14				
5	NESTED LOOPS		50744	12M	
173K (1)	00:00:07				
6	NESTED LOOPS		5891	1052K	
127K (1)	00:00:05				
* 7	HASH JOIN		5781	750K	7048K
104K (1)	00:00:05				
8	TABLE ACCESS FULL	PSP_TAX_COMPANY_SERVICE_INFO	136K	5450K	
488 (1)	00:00:01				
9	TABLE ACCESS FULL	PSP_COMPANY_SERVICE	8915K	782M	
60034 (1)	00:00:03				
* 10	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY	1	50	
4 (0)	00:00:01				
* 11	INDEX RANGE SCAN	SYS_C0020424	1		
2 (0)	00:00:01				
12	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE	9	711	
18 (0)	00:00:01				
* 13	INDEX RANGE SCAN	PSP_EMPLOYEE_FK2	28		
3 (0)	00:00:01				
* 14	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_EMPLOYEE_TAX	1	82	
5 (0)	00:00:01				
* 15	INDEX RANGE SCAN	PSP_EMPLOYEE_TAX_FK2	9		
3 (0)	00:00:01				
* 16	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_TAX_TABLE_MISC_DATA	1	50	
5 (0)	00:00:01				
* 17	INDEX RANGE SCAN	PSP_TAX_TABLE_MISC_DATA_FK1	3		
3 (0)	00:00:01				
18	TABLE ACCESS BY INDEX ROWID BATCHED	PSP_COMPANY_LAW	1	41	
4 (0)	00:00:01				
* 19	INDEX RANGE SCAN	SYS_C0020562	1		
2 (0)	00:00:01				

Peeked Binds (identified by position):

1 - :1 (TIMESTAMP): [Not Printable]

Predicate Information (identified by operation id):

```

7 - access("PCS"."COMPANY_SERVICE_SEQ"="PTCSI"."TAX_COMPANY_SERVICE_INFO_SEQ")
  filter(((("PCS"."STATUS_CD"='ActiveCurrent' AND "PCS"."SERVICE_START_DATE" IS NOT NULL) OR
  (INTERNAL_FUNCTION("PCS"."STATUS_CD") AND "PTCSI"."LAST_QUARTER_TO_FILE"<>0)))
10 - filter(NVL("PC"."IS_DG_DISASSOCIATED",0)=0)
11 - access("PC"."COMPANY_SEQ"="PCS"."COMPANY_FK")
13 - access("PE"."COMPANY_FK"="PC"."COMPANY_SEQ")
14 - filter("PET"."COMPANY_LAW_FK" IS NOT NULL)
15 - access("PET"."EMPLOYEE_FK"="PE"."EMPLOYEE_SEQ")
16 - filter(("MD"."VALUE" IS NOT NULL AND "MD"."MODIFIED_DATE">=:1))
17 - access("MD"."EMPLOYEE_TAX_FK"="PET"."EMPLOYEE_TAX_SEQ")
19 - access("PET"."COMPANY_LAW_FK"="PCL"."COMPANY_LAW_SEQ")

```

Note

- this is an adaptive pla